

STATS 770 - Report

Adam Wong-Toi

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Abstract

The objective of this study is to evaluate the longitudinal association between depression and all-cause mortality by comparing two distinct exposure scores - a PHQ-9 screening threshold (≥ 10) and a stricter DSM-5 diagnostic algorithm. Utilising 2015/2016 NHANES data prospectively linked to the National Death Index through to the end of 2019, multivariable unweighted and complex survey design weighted Cox proportional hazard models were constructed. Models estimated both the total effects and controlled direct effects after adjusting for socio-demographics and selected health behaviours mediators (BMI, sleep, and smoking status). In unweighted models, baseline screening level depression (PHQ-9) was significantly associated with elevated 4-year all-cause mortality. However the stricter clinical DSM-5 metric drastically reduced positive exposures, introducing a severe sample precision penalty. When accounting for complex survey weighting and clustering parameters, structural variance inflation and restricted design degrees of freedom heavily penalised precision, widening confidence intervals and rendering the population level hazard ratios non-significant across both metrics. Lifestyle mediators caused only minor attenuation in hazard ratios. Public health inferences regarding depression attributable mortality are highly sensitive to whether the analytical framework prioritises internal sample precision or population wide external generalisability, highlighting a fundamental methodological trade off between screening sensitivity and clinical diagnostic specificity.

Introduction

The National Health and Nutritional Examination Survey (NHANES), is a survey that collects data about the overall health and well being of Adults and Children in the United States. [1] There are approximately 5000 participants that are involved each year that contribute to the data, to be a representative sample of the wider US population. The survey is extremely comprehensive, collecting over 1500 variables across a range of tables and questionnaires [2]. This creates an extremely rich source of data that runs on 2-year cycles for analysis. Additionally, it is possible to link this data with other data sets, including the National Death Index (NDI).

The NHANES is a regularly repeating stratified multi-stage cross-sectional study. Every survey does not have the same group of individuals in it, as they are sampled, stratified and weighted accordingly for each cycle of the study. There are typically no repeated participants across different cycles. The study ensure that those individuals that are in smaller demographics are sampled accordingly to get accurate measurements across all demographic classes and variables, as well as appropriate stratification and weighting from the survey units (typically 15 different counties in the USA) [1].

For this analysis, I focus on a small number of variables and possible predictors from the 2015-2016 NHANES survey, linked with the 2019 NDI data. This allows for approximately 3-4 year follow-up period for mortality related indicators.

Research Questions of Interest

For this analysis, I am interested in answering the following research questions:

1. How is baseline probable depression ($\text{PHQ-9} \geq 10$) risk of long-term all-cause mortality?
2. Does a strict DSM-5 algorithmic definition of depression reveal a different mortality risk profile than the standard PHQ-9 screening score?

Statistical Analysis

As there were so many possible variables that could be included in the analysis, it was quite difficult to narrow this down to a sensible number of possible predictors. I chose what I believe to be the most relevant demographic data, as well as common possible covariates for the analysis. These are summarised in Table 1 and 2. There were a total of 9971 individuals from the study period, but after including only those that were eligible for linkage with the National Death Index (i.e no children/under-18s or ineligible due to privacy reasons), there were 5974 individuals remaining for the primary analysis. As part of the data cleaning process, some covariates were adjusted/edited, to make more sensible groupings. For example, making composite variables of smokers, to be NeverSmoked, FormerSmoker, CurrentSmoker, based on data about if individuals were a current smoker, had ever smoked (but not a current smoker) or never smoked. For full information about the data cleaning, please see the appendix.

Assessment of Missing Data

The dataset was not complete, with some variables containing up to 60% missing data. The covariates that had the most missing data were the smoking age, alcohol and drug related covariates - which were optional as part of the NHANES survey. 14.2% of the data was missing complete information to construct the different depression scores. All other covariates had more than 90% complete data. Individuals that did not have the depression related scores were automatically removed from the modelling when this was conducted. All other individuals were retained to begin with. However, after an initial analysis, the significant lack of data in the drugs/smoking age/alcohol variables, reduced the valid mortality individuals in the model to less than 20 individuals. These models that included these covariates were not practical or statistically sensible. These covariates were then removed from the models. All other covariates and confounders were retained. Note, that although age started smoking was removed, there was still a valid smoking status covariate that was not removed from the data, and had only 0.18% missingness. Based on this, I felt that it was sufficient to conduct this analysis without doing some formal procedure to replace the missingness (like Baseline or Final Value Carried Forward or Multiple Imputation).

Depression Scores

The PHQ-9 score is a standardised questionnaire use to self-test and screen for possible depression, and measure its severity. Each of the 9 indicators can be scored 0 to 3 with the scores representing the frequency from 0: not at all, to 3: nearly every day.

The maximum possible score for this indicator is 27, with common thresholds set at 0, 5, 10, 15 and 20 to represent no, mild, moderate, moderately severe and severe depression.[3] The choice to create a binary indicator for ($\text{PHQ} \geq 10$), represented those individuals with moderate or higher scores for depression [3].

The DSM-5 algorithmic definition of depression uses the same PHQ-9 scores, but has a more complex algorithm to define possible depression. Candidates must have scored a 2, or 3 in at least 5 across the 9 questions, as well as at least a 2 in either the first or second question. (Depressed mood, or little interest in doing anything) [4].

As both of these depression scores are supported by literature of defining and utilising these measures, I opted to keep this analysis consistent with these scoring metrics.

Hypotheses

Based on my variables, I formed the following hypotheses about the variables to create my causal diagram prior to model fitting. Given the scope and nature of these variables, these may not be a comprehensive list of valid causal pathways, but form the basis for my analysis, and are what I believe are the most important. Mortality was assumed to be from any cause, not specific causes - hence all-cause mortality.

- Depression impacts all-cause mortality, sleep duration, BMI, smoking status, drug and both forms of alcohol use.
- Age impacts depression, all-cause mortality, sleep duration and BMI.
- Sex impacts depression, all-cause mortality, smoking status, alcohol and both forms of drug use.
- Ethnicity & marital status impacts depression and all-cause mortality.
- All-cause mortality is impacted by sleep duration, BMI, smoking status, alcohol and both forms of drug use

As mentioned as part of the missing data assessment, the age started smoking, alcohol and drug use related covariates were removed to yield the causal diagram - simplified to remove these covariates in Figure 1.

Directed Acyclic Graph (DAG) of Variable Pathways

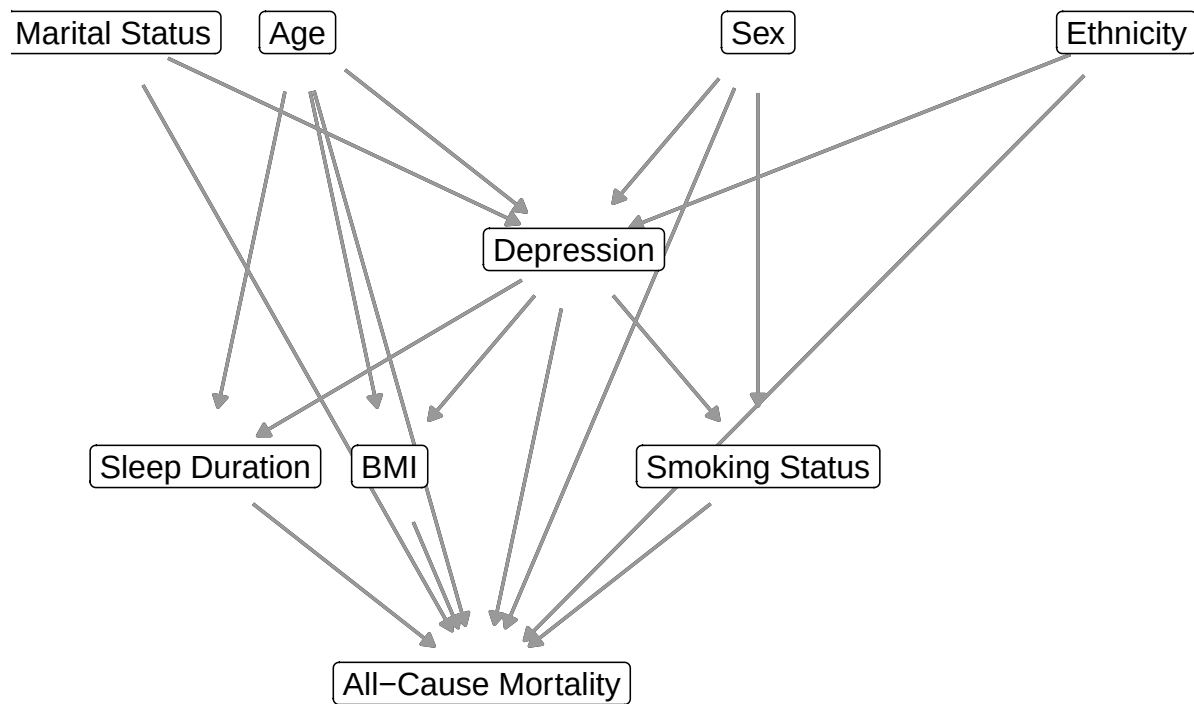


Figure 1: Causal Diagram for Simplified Effects of Depression on All Cause Mortality

Statistical Models

Survival analysis was conducted using Cox proportional hazard regression models to estimate hazard ratios and confidence intervals for all-cause mortality. The time-to-event scale was defined in person months of follow-up from the baseline interview data to the documented date of death, or censoring on December 31,

2019 (the end of the publicly linkable National Death Index). Assumptions of proportional hazard was tested for all models fit, and if not stated otherwise, the models met these assumptions.

Models were fit for the raw, total and direct effect, with the latter two also having models that accounted for the complex multi-stage cluster survey design of the NHANES study, using the `survey` package. The primary sampling units, geographical sampling strata and baseline 2 year Mobile Examination Centre weights were utilised, as defined in the NHANES demographic data.

To account for possible ‘lonely Primary Sampling Units’ where only single individuals remained after data cleaning, the models were set to adjust the variance by setting any singletons at the population grand mean to prevent standard error underestimation. For the total effect, with survey weights/strata adjustment, parameter inflation did have to be adjusted by collapsing the marriage status variable into a 2 level categorical variable instead of a 6-level variable, to ensure structural matrix inversion was possible without leading to singularity in the model fitting.

All analyses were performed in R, version 4.5.3 (R Foundation for Statistical Computing). Significance levels were 0.05 (α) and all statistical tests were two-sided.

Results

Baseline characteristics are shown in Table 1 for the core demographic variables for both metrics of depression score. Additionally, a table of possible variables that are either confounders or mediators that are considered to be added to models for adjustments to get direct/total effects is shown in Table 2.

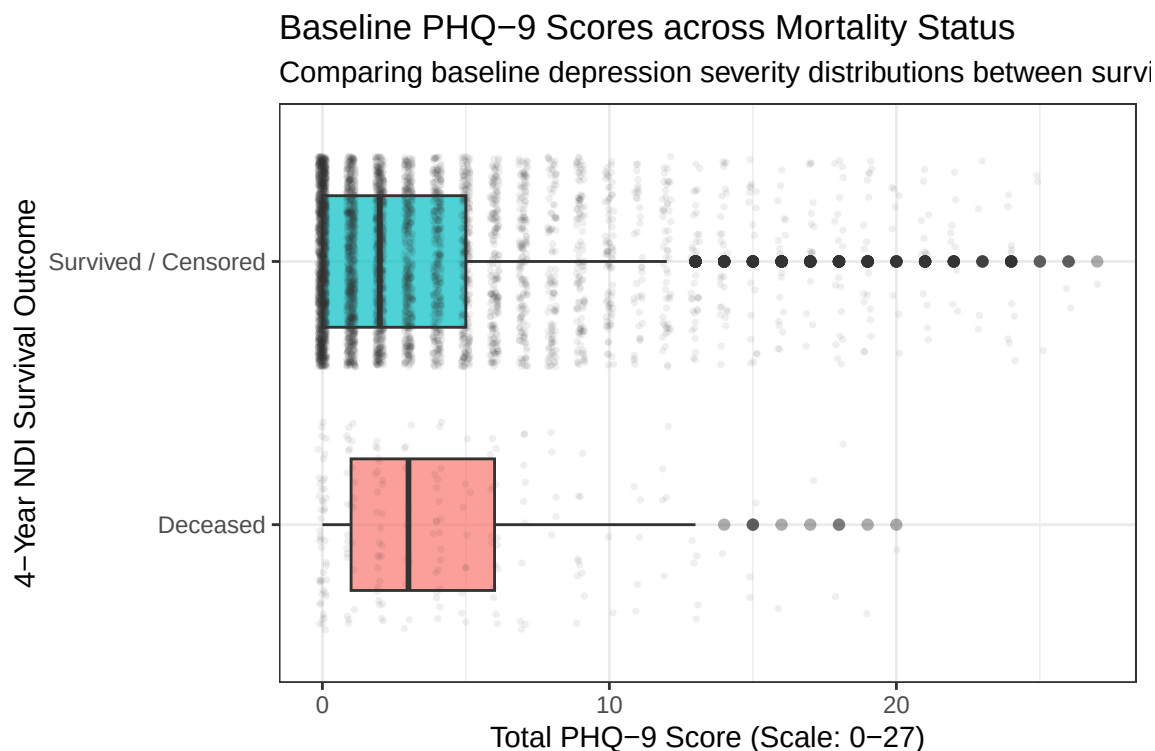


Figure 2: Baseline PHQ-9 Scores across mortality status

Table 1: Socio-Demographic Characteristics and Survival Outcomes Stratified by Two Depression Classifications

Characteristic	Probable Depression (PHQ-9 10+)			DSM-5 Algorithmic Depression		
	No N = 4,709	Yes N = 415	p-value	No N = 4,931	Yes N = 193	p-value
Age in years at screening	48.2 (18.5)	49.1 (18.0)	0.4	48.2 (18.5)	49.8 (17.7)	0.2
Sex			<0.001			0.015
Male	2,339 (50%)	164 (40%)		2,426 (49%)	77 (40%)	
Female	2,370 (50%)	251 (60%)		2,505 (51%)	116 (60%)	
Race/Ethnicity			<0.001			0.006
MexicanAmerican	861 (18%)	69 (17%)		895 (18%)	35 (18%)	
OtherHispanic	586 (12%)	70 (17%)		619 (13%)	37 (19%)	
Non-HispWhite	1,557 (33%)	152 (37%)		1,641 (33%)	68 (35%)	
Non-HispBlack	1,010 (21%)	84 (20%)		1,055 (21%)	39 (20%)	
Non-HispAsian	521 (11%)	18 (4.3%)		532 (11%)	7 (3.6%)	
Other/Multi-Racial	174 (3.7%)	22 (5.3%)		189 (3.8%)	7 (3.6%)	
Marital Status			<0.001			<0.001
Divorced	465 (10%)	64 (16%)		498 (11%)	31 (17%)	
LivingWithPartner	435 (9.7%)	39 (9.8%)		451 (9.6%)	23 (12%)	
Married	2,320 (52%)	133 (33%)		2,398 (51%)	55 (30%)	
NeverMarried	794 (18%)	102 (26%)		848 (18%)	48 (26%)	
Separated	145 (3.2%)	28 (7.0%)		164 (3.5%)	9 (4.9%)	
Widowed	320 (7.1%)	32 (8.0%)		333 (7.1%)	19 (10%)	
Missing	230	17		239	8	
Mortality Status	183 (3.9%)	27 (6.5%)	0.015	198 (4.0%)	12 (6.2%)	0.13
Follow-up (Months)	47.6 (8.3)	46.6 (9.8)	0.047	47.5 (8.4)	46.6 (8.8)	0.2

¹ Mean (SD); n (%)

² Welch Two Sample t-test; Pearson's Chi-squared test with simulated p-value (based on 2000 replicates)

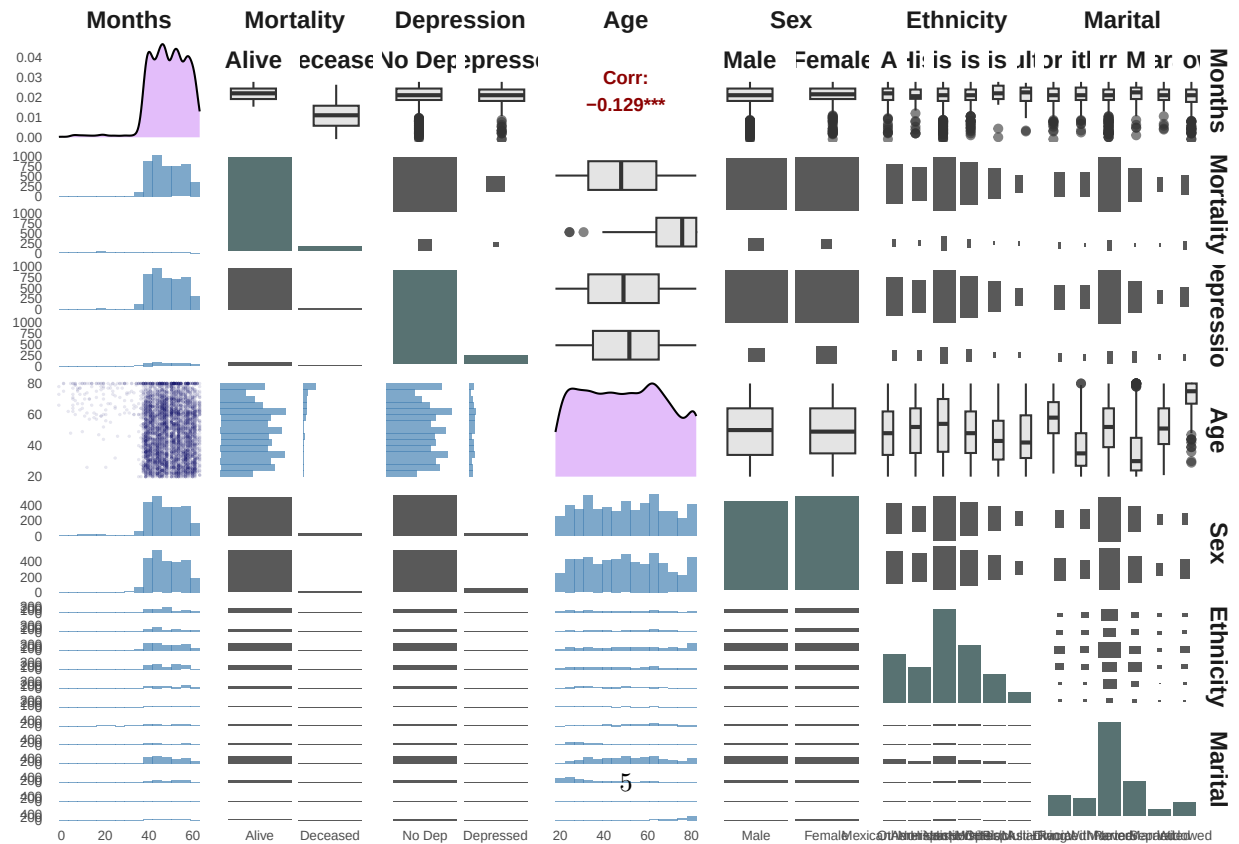


Figure 3: Pairs Plot of Demographic Covariates

Table 2: Clinical Baseline Confounders and Substance Use Behaviours Stratified by Two Depression Classifications

Characteristic	Probable Depression (PHQ-9 10+)			DSM-5 Algorithmic Depression		
	No N = 4,709	Yes N = 415	p-value	No N = 4,931	Yes N = 193	p-value
Body Mass Index (kg/m**2)	29.4 (7.0)	31.2 (8.4)	<0.001	29.4 (7.1)	31.3 (8.0)	0.001
Systolic: Blood pres (1st rdg) mm Hg	125.1 (18.5)	125.5 (17.4)	0.7	125.1 (18.4)	126.9 (18.9)	0.2
Total Cholesterol (mg/dL)	188.9 (41.5)	191.0 (43.5)	0.3	189.0 (41.6)	190.6 (42.6)	0.6
Sleep hours	7.7 (1.5)	7.9 (2.1)	0.2	7.7 (1.5)	8.1 (2.2)	0.045
Sleep Category			<0.001			<0.001
Short (<6 hrs)	359 (7.7%)	59 (14%)		393 (8.0%)	25 (13%)	
Normal (6-8 hrs)	2,833 (60%)	192 (47%)		2,942 (60%)	83 (44%)	
Long (>8 hrs)	1,497 (32%)	160 (39%)		1,576 (32%)	81 (43%)	
Age started smoking cigarettes regularly	22.4 (66.5)	21.6 (62.8)	0.9	22.6 (68.1)	17.4 (6.9)	0.002
Smoking Status			<0.001			<0.001
NeverSmoked	2,822 (60%)	168 (40%)		2,923 (59%)	67 (35%)	
FormerSmoker	1,079 (23%)	88 (21%)		1,124 (23%)	43 (22%)	
CurrentSmoker	802 (17%)	159 (38%)		878 (18%)	83 (43%)	
Alcohol Drinks/Day	2.7 (2.3)	3.3 (3.1)	<0.001	2.7 (2.3)	3.8 (3.5)	<0.001
History of Marijuana Use			<0.001			<0.001
No	1,600 (51%)	90 (33%)		1,657 (51%)	33 (26%)	
Yes	1,527 (49%)	185 (67%)		1,619 (49%)	93 (74%)	
History of Hard Drug Use			<0.001			0.002
No	3,287 (85%)	266 (76%)		3,431 (85%)	122 (75%)	
Yes	586 (15%)	82 (24%)		627 (15%)	41 (25%)	

¹ Mean (SD); n (%)

² Welch Two Sample t-test; Pearson's Chi-squared test with simulated p-value (based on 2000 replicates)

Given the large number of possible covariates, there are a significant number of possible pair wise combinations of variables to visualise. I attempted to visualise a pairs plot of these variables, however, given the extensive nature of the data and the number of variables, the plot is not overly helpful. For this, please see the appendix. However, a pairs plot of the demographic variables is shown in Figure 3. This is still reasonably difficult to understand interpret given the number of variables, hence why it is easier to interpret and assess these variables in a table due to the complexity of the data set.

PDQ-9 $\geq$ 10 indicator

The unadjusted baseline model, individuals that met the screening threshold had a 1.71-fold higher risk of all-cause mortality during the followup period compared to those below the threshold (Hazard Ratio (HR): 1.71, 95% CI: 1.14 – 2.57, p-value: 0.009).

To isolate the total causal effect of depression on mortality, the model was adjusted for baseline demographic confounders include age, sex, ethnicity, and marital status. Accounting for these structural confounders, slightly strengthened the magnitude of the risk association (HR: 1.81, 95% CI: 1.2 – 2.73, p-value: 0.005).

Finally, to compute the controlled direct effect of baseline depression, independent of post-exposure behavioural pathways, the model was adjusted for lifestyle mediators: smoking status, BMI and average nightly sleep duration. While the inclusion of these covariates resulted in a minor attenuation, the direct hazard remained elevated and highly statistically significant (HR: 1.67, 95% CI: 1.09 – 2.55, p-value: 0.017). This indicated that a substantial proportion of the mortality burden associated with a positive PHQ-9 screening operates through pathways independent of sleep, smoking status and BMI.

DSM-5 Indicator

To address the second research question, I replaced the broader PDQ-9 binary indicator with the stricter, symptom-pattern matching algorithm modelled directly on the DSM-5 criteria for Major Depressive Disorder (requiring at least 5 concurrent symptoms, inclusive of a verified core indicator of either a depressed mood or apathy/loss of interest).

By shifting to this stricter definition altered the statistical significance of the models, though the estimate remains moderately elevated. In the unadjusted model individuals meeting the DSM-5 algorithmic criteria exhibited a 1.58-fold higher hazard of all-cause mortality compared to those that did not meet the DSM-5 criteria, however this was a non-significant relationship. (HR: 1.58, 95% CI: 0.88 – 2.83, p-value: 0.122).

Controlling for demographic variables within the total direct effect, results in an estimate similar to that of the unadjusted model (HR: 1.54, 95% CI: 0.85 – 2.78, p-value: 0.153), but remained statistically insignificant. Same with the direct effect through the mediators (HR: 1.45, 95% CI: 0.8 – 2.64, p-value: 0.221), which also remained insignificant.

The results of the PHQ-9 and DSM-5 Depression Total Effect survival curve models can be seen in Figure 3 and 4. These are the survival curves for just the two Total Effects. Curves for the raw and direct effect are not plotted due to similarities with Figure 3 and 4, to save space.

Adjusted Survival Trajectories by Baseline Depression Status

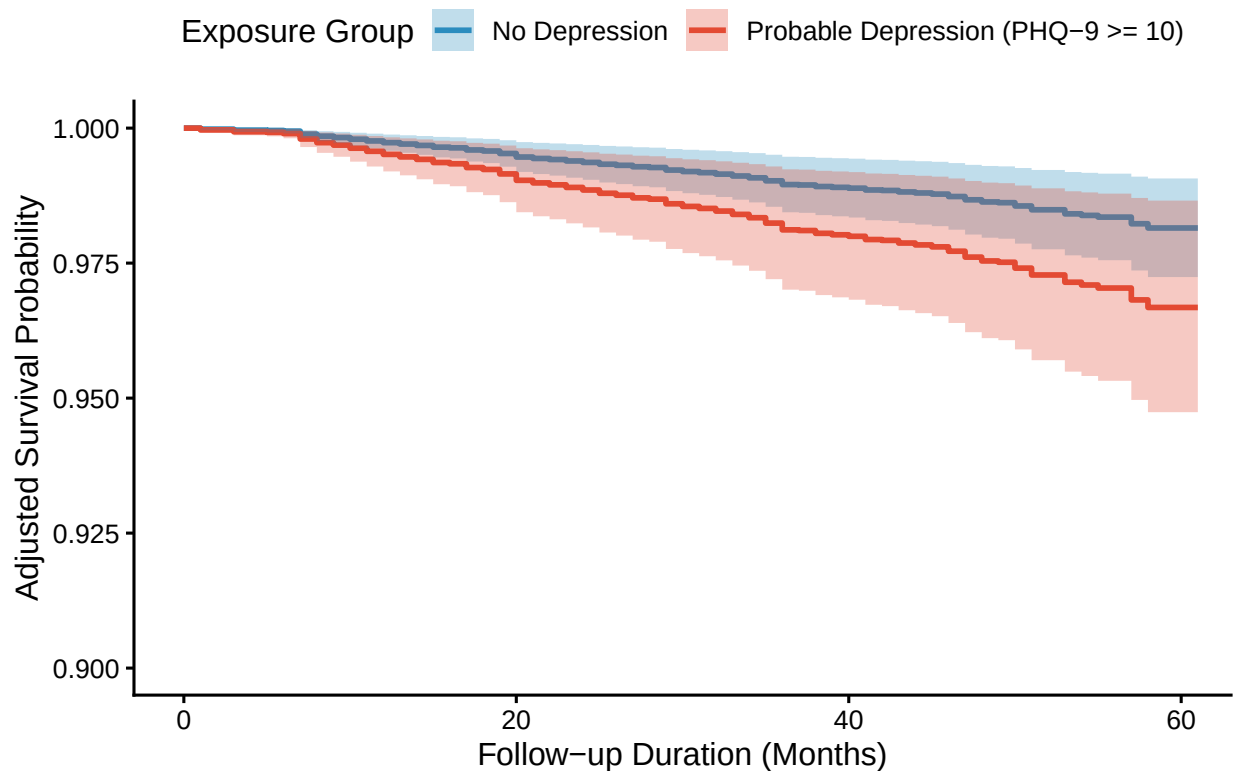


Figure 4: Survival Curve for Total Effect Model for Probable PHQ-9 Depression

Adjusted Survival Trajectories by Baseline Depression Status

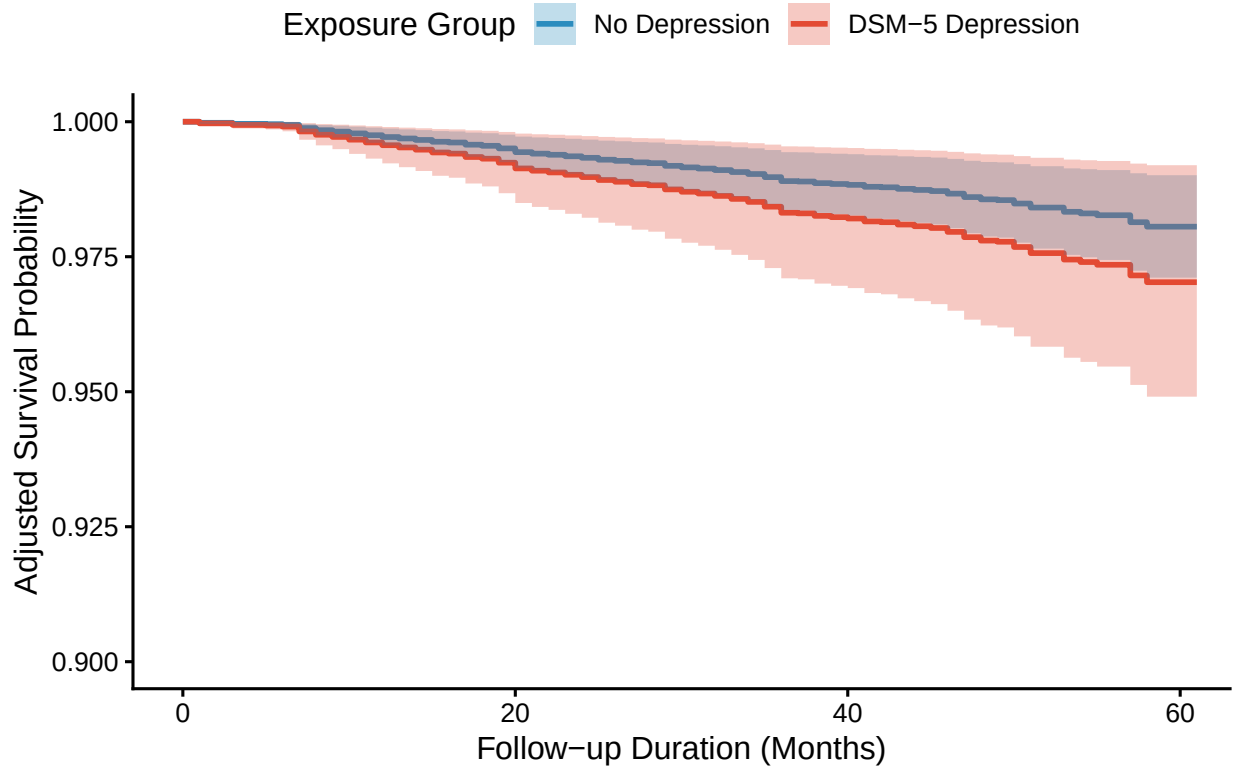


Figure 5: Survival Curve for Total Effect Model for DSM-5 Depression

Sensitivity Analysis: Complex Survey Design and Sampling Weights

To address the generalisability of the findings to the broader United States population, a final analysis was conducted. This utilised the survey design and sampling weights from the NHANES data to explicitly adjust for the primary sampling unit, clustering, demographic strata and sampling weights of the data.

The integration of the complex survey design parameters introduced a distinct divergence in our causal estimates and distribution of standard errors compared to the un-weighted framework. For the screening level PHQ-9 indicator, accounting for survey weights shifted the baseline point estimates up. In the survey weighted unadjusted framework, the hazard ratio for individuals with baseline probable depression rose to 1.82 (95% CI: 1.08 – 3.08, p-value: 0.025). This upward translation persisted into the survey-weighted total effect model which exhibited the most pronounced association observed for the screening metric. When generalised to the target population, baseline probable depression effectively doubled the four-year risk of all cause-mortality (HR: 2.01, 95% CI: 1.1 – 3.65, p-value: 0.022).

Crucially, adjusting for the lifestyle mediators within the survey-weighted direct effect model resulted in minor attenuation, like with the un-weighted model (HR: 1.83, 95% CI: 0.98 – 3.4, p-value: 0.057). Due to the complex design, which corrects for clustering and stratification, the robust standard error increased to 0.317. This widening of the standard error produced a wider confidence interval, and thus, the direct effect of baseline screening-level depression on mortality just misses the conventional significance threshold when generalised from the sample to the target US population.

For the stricter DSM-5 metric, projecting the findings onto the target US population, generated a substantial upward shift in the effect sizes, but this was accompanied by an extreme inflation of the standard errors that rendered all clinical specifications non-significant.

- Unadjusted raw effect Hazard Ratio: 1.99 (95% CI: 0.79 – 5.04, p-value: 0.146).
- Total effect Hazard Ratio: 2.15 (95% CI: 0.78 – 5.92, p-value: 0.137).
- Direct effect Hazard Ratio: : 1.88 (95% CI: 0.67 – 5.25, p-value: 0.227).

The full set of results from each of the models and depression indicators can be found in Table 3 and 4, as well as in Figure 5. Survival curves for these models were not generated due to both space saving and the fact that they were non significant differences across all models for both depression scores when weighted by survey design.

Comparing these outcomes across both indicators highlights a critical analytical trade off. While the stricter DSM-5 categorisation consistently yielded higher point estimates of mortality risk in the true US population compared to its un-weighted counterparts, the highly specific clinical criteria vastly restricted the subpopulation size. When overlaid into the complex survey design and sampling weights, this sample restriction resulted in a severe deficit of statistical precision, producing wide confidence intervals that failed to establish a definitive statistical association across any of the model specifications.

Table 3: Unweighted model results across both depression indicators.

Depression Metric	Adjustment Level	HR (95% CI)	p-value
PHQ-9 Depression	Unadjusted (Raw)	1.71 (1.14, 2.57)	0.009
PHQ-9 Depression	Total Effect	1.81 (1.20, 2.73)	0.005
PHQ-9 Depression	Direct Effect	1.67 (1.09, 2.55)	0.017
DSM-5 Depression	Unadjusted (Raw)	1.58 (0.88, 2.83)	0.122
DSM-5 Depression	Total Effect	1.54 (0.85, 2.78)	0.153
DSM-5 Depression	Direct Effect	1.45 (0.80, 2.64)	0.221

Table 4: Survey weighted model results across both depression indicators.

Depression Metric	Adjustment Level	HR (95% CI)	p-value
PHQ-9 Depression	Unadjusted (Raw)	1.82 (1.08, 3.08)	0.025
PHQ-9 Depression	Total Effect	2.01 (1.10, 3.65)	0.022
PHQ-9 Depression	Direct Effect	1.83 (0.98, 3.40)	0.057
DSM-5 Depression	Unadjusted (Raw)	1.99 (0.79, 5.04)	0.146
DSM-5 Depression	Total Effect	2.15 (0.78, 5.92)	0.137
DSM-5 Depression	Direct Effect	1.88 (0.67, 5.25)	0.227

Discussion

To evaluate the overall methodology in this report, there are a few considerations to bear in mind.

Classification of Depression

The $\text{PHQ-9} \geq 10$ approach may capture current elevated depression symptoms. It is shown to be an effective screening method, but could introduce classification bias because it captures temporary situational distress

or grief, alongside clinical depression. The DSM-5 criteria enforces a rigid structural definition that has a higher threshold than the PHQ-9 score, which is more closely associated with Major Depressive Disorder than a possible depression indicator from the PHQ-9 score.

Additionally, the stricter DSM-5 score reduces the sample size of positive exposures in the sample. When stratified by this metric, the sample size has reduced, leading to wider confidence intervals and loss of statistical significance in the more complex survey weight and sampling design models.

Timeline of followup

The 4-year window for follow-up may be too short for mortality tracking. Depression or depressive disorders may have longer term effects on humans over a longer period of time, like decades, not just a 4 year window. The long term risk may be under-represented by this shorter time frame. Additionally, larger sample sizes of mortality would likely occur from longer follow-up windows. This impact could result in protopathic bias (reverse causality).

This is one of the limitations of this study, and the choice to use the NHANES data for this analysis. The National Death Index public linking of data is only available until the end of 2019. It is possible to use older NHANES data, but I did not want to use data that was too old, due to possible changes in data storage, or incompatibilities of data linkage. Whilst it would be possible to use older NHANES data and link this to the NDI data to December 2019, the choice to select 2015/2016 data was just by personal preference, but in hindsight could have been 1 or 2 cycles earlier to get a longer follow-up time.

The Cox Proportional Hazard models are able to handle the right-censoring effectively. Individuals who were still alive at the end of the study period, will contribute person time, right up to the point of censoring. Part of one of the limitations of this model is that using the baseline measurement for depression scores from the NHANES is at the baseline, there is no further followup on the participants, that there is only the ability to have the baseline depression score factored into the model. As a result of this, there is no way to know when the depression started. Additionally, the all-cause death indicator, may not be depression related - which is a key limitation of this study.

A major strength to this study design, is that the data is linked with the NDI. Unlike traditional cohort studies which usually suffer from participant drop out or attrition bias, the linking with the NDI minimised this loss to followup.

Missing Data / Possible Confounders

Another limitation of the study is that the possible confounders identified with age started smoking, alcohol and drug use, which were removed due to high missingness, could have an impact on the depression scores. The results of these parts of the NHANES study being optional to answer, or people refusing to answer questions relating to possibly illegal behaviours, could impact why there is significant missingness in these covariates. These confounders may be significant indicators/mediators to the overall depression to all-cause mortality. So although this analysis did not include these due to significant missingness, future work could explore possible imputation of this data, or aggregation of reliable sources to consider merging with this analysis. This has introduced non-response bias from the optional survey sections relating to alcohol/drug use.

More Complex Survey Designs

With the implementation of the more complex survey weighted design to this analysis, we saw that the point estimates increased when the weighting was considered. This shows that the association between depression and mortality is stronger in the specific demographic strata that the NHANES samples from. A cost of this, is that there is more uncertainty (larger standard errors). This model fit has penalised the fact that our data is not perfectly independent, and the population level statistical hazard has lost its significance.

Possible improvements to this would be either reducing model complexity, or employing a longer followup time, or possible aggregation of samples across survey periods to increase the sample size for analysis. This aggregation of multiple NHANES may pose issues with differing followup times, but with careful selection, and data processing, may be possible to strengthen the relationships with more complex survey designs. This was not conducted as part of this analysis.

This more complex survey design, taken only for one NHANES cycle has also restricted the degrees of freedom (The number of clusters minus the number of strata). This restriction is why adjustments had to be made to the marriage status variable, to reduce the number of predictors being estimated, to be less than the degrees of freedom, to generate a valid model estimate. By using a more complex model, with quite a lot of predictors, as well as only one cycle, the degrees of freedom are used up quickly, generating a model that has naturally inflated variance (from fewer degrees of freedom), penalising the precision of the model estimates at the population level.

Sensibility of the Study Design/Analysis

Whilst a prospective cohort study over many years may provide more accurate information of when depression indicators change in individuals and track them for longer periods of time, could result in a more accurate definitive answer of the questions of interest for this study, this would be a lot more expensive and administratively burdening to collect and store this data. I feel that the NHANES study data, whilst not perfect, provides an efficient way to conduct this analysis. Especially being able to link the NHANES with the NDI, there is a sound and equally valid methodology for this analysis.

Summary

This study highlights a notable trade-off between screening sensitivity and diagnostic specificity when evaluating the association between depression and all cause mortality using NHANES data. While the unweighted sample models demonstrate a statistically significant internal association for the broader PHQ-9 criteria, moving to the stricter DSM-5 criteria drastically restricts the sample size. When accounting for the complex survey design parameters, the sample restriction results in a severe penalty in precision, inflating standard errors and rendering the population level estimates non-significant. Despite key limitations including a truncated 4-year followup window, with possible selection bias and missing behavioural confounders, the passive longitudinal linkage to the National Death Index provides a highly sensible, resource efficient framework for tracking survival and hazard. Ultimately these findings reveal that public health conclusions regarding depression related mortality are highly sensitive to whether the analytical framework prioritises internal sample precision or broad, population level, external generalisability.

References

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- [2] - Ale, L., Gentleman, R., Sonmez, T. F., Sarkar, D., & Endres, C. (2024). nhanesA: achieving transparency and reproducibility in NHANES research. Database, 2024. <https://doi.org/10.1093/database/baae028>
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Appendix

Data loading

```
# Data read in

# Demographic Data
demo_df <- read_xpt("datasets/DEMO_I.XPT") |>
  select(SEQN, RIAGENDR, RIDAGEYR, RIDRETH3, DMDMARTL, WTMEC2YR, SDMVPSU, SDMVSTRA)

# Mortality Data
mort_df <- read_fwf(
  "datasets/NHANES_2015_2016_MORT_2019_PUBLIC.dat",
  fwf_positions(
    start = c(1, 15, 16, 17, 43), # Precision shift: permth_int is at character 43
    end   = c(6, 15, 16, 19, 45),
    col_names = c("SEQN", "eligstat", "mortstat", "ucod_leading", "permth_int")
  ),
  col_types = cols(
    SEQN          = col_double(),
    eligstat      = col_double(), # 1 = Eligible, 2 = Child, 3 = Ineligible.
    mortstat      = col_double(), # Mortality status
    ucod_leading  = col_double(), # Underlying cause of death 1-Stroke/2-Cancer/3-Respiratory/4-Stroke/5-
    permth_int    = col_double()  # Person Months of Followup
  )
)
```

```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
# Depression Score
dpq_df <- read_xpt("datasets/DPQ_I.XPT") |>
  select(SEQN, starts_with("DPQ"))

# Other possible covariates
tchol_df    <- read_xpt("datasets/TCHOL_I.xpt") |> select(SEQN, LBXTC) # Total cholesterol
hdlchol_df  <- read_xpt("datasets/HDL_I.xpt")   |> select(SEQN, LBDHDD) # HDL Cholesterol
bloodp_df   <- read_xpt("datasets/BPX_I.xpt")   |> select(SEQN, BPXSY1, BPXDI1) # Systolic and Dias
bmi_df      <- read_xpt("datasets/BMX_I.xpt")   |> select(SEQN, BMXBMI, BMXHT, BMXWT) # BMI, Standi
sleep_df    <- read_xpt("datasets/SLQ_I.xpt")   |> select(SEQN, SLD012, SLQ300, SLQ310) # Sleep Hrs,
smoke_df    <- read_xpt("datasets/SMQ_I.xpt")   |> select(SEQN, SMQ020, SMD030, SMQ040) # Smoked 100
alc_df      <- read_xpt("datasets/ALQ_I.xpt")   |> select(SEQN, ALQ130) # Avg Alcoholic drinks per d
drug_df     <- read_xpt("datasets/DUQ_I.xpt")   |> select(SEQN, DUQ200, DUQ240) # Marijuana, Meth/Co
```

Merging Data

```
# Joining Data
```

```

# Base cohort - eligible for linkage with NDI
base_cohort <- demo_df |>
  inner_join(mort_df, by = "SEQN") |>
  filter(RIDAGEYR >= 18, eligstat == 1)

# Joining other data sets - left join does not delete participants
analysis_df <- base_cohort |>
  left_join(dpq_df, by = "SEQN") |>
  left_join(tchol_df, by = "SEQN") |>
  left_join(hdlchol_df, by = "SEQN") |>
  left_join(bloodp_df, by = "SEQN") |>
  left_join(bmi_df, by = "SEQN") |>
  left_join(sleep_df, by = "SEQN") |>
  left_join(smoke_df, by = "SEQN") |>
  left_join(alc_df, by = "SEQN") |>
  left_join(drug_df, by = "SEQN")

```

Data Cleaning

```

analysis_clean_df <- analysis_df |>
  # -----
  # Clean NHANES Missing/Refused Codes (Convert to true NA)
  # -----
  mutate(
    # DPQ questionnaire missing values (7 = Refused, 9 = Don't Know)
    across(starts_with("DPQ"), ~ ifelse(.x %in% c(7, 9), NA_real_, .x)),
    # General missing values across survey answers
    DMDMARTL = ifelse(DMDMARTL %in% c(77, 99), NA_real_, DMDMARTL),
    SMQ020 = ifelse(SMQ020 %in% c(7, 9), NA_real_, SMQ020),
    SMQ040 = ifelse(SMQ040 %in% c(7, 9), NA_real_, SMQ040),
    ALQ130 = ifelse(ALQ130 %in% c(777, 999), NA_real_, ALQ130),
    DUQ200 = ifelse(DUQ200 %in% c(7, 9), NA_real_, DUQ200),
    DUQ240 = ifelse(DUQ240 %in% c(7, 9), NA_real_, DUQ240),

    # Recoding factor levels here so they are safely masked inside mutate
    sex = case_when(
      RIAGENDR == 1 ~ "Male",
      RIAGENDR == 2 ~ "Female",
      TRUE ~ NA_character_
    ) |> factor(levels = c("Male", "Female")),

    race_ethnicity = case_when(
      RIDRETH3 == 1 ~ "MexicanAmerican",
      RIDRETH3 == 2 ~ "OtherHispanic",
      RIDRETH3 == 3 ~ "Non-HispWhite",
      RIDRETH3 == 4 ~ "Non-HispBlack",
      RIDRETH3 == 6 ~ "Non-HispAsian",
      RIDRETH3 == 7 ~ "Other/Multi-Racial",
      TRUE ~ NA_character_
    ) |> factor(levels = c("MexicanAmerican", "OtherHispanic", "Non-HispWhite",

```

```

        "Non-HispBlack", "Non-HispAsian", "Other/Multi-Racial")),

marital_status = case_when(
  DMDMARTL == 1 ~ "Married",
  DMDMARTL == 2 ~ "Widowed",
  DMDMARTL == 3 ~ "Divorced",
  DMDMARTL == 4 ~ "Separated",
  DMDMARTL == 5 ~ "NeverMarried",
  DMDMARTL == 6 ~ "LivingWithPartner",
  TRUE ~ NA_character_
) |> factor(),

cause_of_death = case_when(
  mortstat == 0 ~ "Survived/Censored",
  ucod_leading == 1 ~ "CardiovascularDisease",
  ucod_leading == 2 ~ "Cancer",
  ucod_leading == 10 ~ "Other",
  TRUE ~ NA_character_
) |> factor(levels = c("Survived/Censored", "CardiovascularDisease",
                        "Cancer", "Other")),
# NOTE: Chronic Respiratory, Stroke and Accident categories were removed as no
# Individuals had this cause of death

smoking_status = case_when(
  SMQ020 == 2 ~ "NeverSmoked", #never smoked
  SMQ020 == 1 & SMQ040 == 3 ~ "FormerSmoker", #Smoked at least 100 cigarettes, but not a current sm
  SMQ020 == 1 & SMQ040 %in% c(1, 2) ~ "CurrentSmoker", #Smoked at least 100 cigarettes, and sometim
  TRUE ~ NA_character_
) |> factor(levels = c("NeverSmoked", "FormerSmoker", "CurrentSmoker")),

# Clean the smoking age placeholder codes (77777 / 99999) to true NA
age_started_smoking = ifelse(SMD030 %in% c(777, 999, 77777, 99999), NA_real_, SMD030),

used_marijuana = case_when(
  DUQ200 == 1 ~ "Yes",
  DUQ200 == 2 ~ "No",
  TRUE ~ NA_character_
) |> factor(levels = c("No", "Yes")),

used_hard_drugs = case_when(
  DUQ240 == 1 ~ "Yes",
  DUQ240 == 2 ~ "No",
  TRUE ~ NA_character_
) |> factor(levels = c("No", "Yes")),

sleep_category = case_when(
  SLD012 < 6 ~ "Short (<6 hrs)",
  SLD012 >= 6 & SLD012 <= 8 ~ "Normal (6-8 hrs)",
  SLD012 > 8 ~ "Long (>8 hrs)",
  TRUE ~ NA_character_
) |> factor(levels = c("Short (<6 hrs)", "Normal (6-8 hrs)", "Long (>8 hrs)")),

```

```

# -----
#  DSM Depression Algorithm Criteria
# -----
# Core items 1-8 require a severity of >= 2 (More than half the days).
# Suicidal ideation (item 9) counts if it is experienced at all (>= 1).
dsm_core_1 = ifelse(DPQ010 >= 2, 1, 0),
dsm_core_2 = ifelse(DPQ020 >= 2, 1, 0),
dsm_sym_3  = ifelse(DPQ030 >= 2, 1, 0),
dsm_sym_4  = ifelse(DPQ040 >= 2, 1, 0),
dsm_sym_5  = ifelse(DPQ050 >= 2, 1, 0),
dsm_sym_6  = ifelse(DPQ060 >= 2, 1, 0),
dsm_sym_7  = ifelse(DPQ070 >= 2, 1, 0),
dsm_sym_8  = ifelse(DPQ080 >= 2, 1, 0),
dsm_sym_9  = ifelse(DPQ090 >= 1, 1, 0), # Suicidal tendency

# Sum up how many distinct DSM symptoms are tracking positive
dsm_symptom_count = rowSums(across(c(dsm_core_1:dsm_sym_9)), na.rm = FALSE),

# DSM-5 diagnosis requires >= 5 total symptoms AND at least one must be Core 1 or Core 2
dsm5_mdd = case_when(
  dsm_symptom_count >= 5 & (dsm_core_1 == 1 | dsm_core_2 == 1) ~ 1,
  is.na(dsm_symptom_count) ~ NA_real_,
  TRUE ~ 0
),

# -----
#  Creating Composite Depression Score
# -----

# PHQ-9 Probably Depression Sum Score (Requires all 9 core questions)
phq9_sum = rowSums(across(DPQ010:DPQ090), na.rm = FALSE),
probable_depression = ifelse(phq9_sum >= 10, 1, 0),

depression_5bin = case_when(
  phq9_sum >= 0 & phq9_sum <= 4 ~ "None-Minimal",
  phq9_sum >= 5 & phq9_sum <= 9 ~ "Mild",
  phq9_sum >= 10 & phq9_sum <= 14 ~ "Moderate",
  phq9_sum >= 15 & phq9_sum <= 19 ~ "Moderately Severe",
  phq9_sum >= 20 & phq9_sum <= 27 ~ "Severe",
  TRUE ~ NA_character_
) |> factor(levels = c("None-Minimal", "Mild", "Moderate",
  "Moderately Severe", "Severe"))

) |> # End of mutate for all new variables
# -----
# Structural Rename and Factor Label Mapping
# -----
select(
  # Core Keys and Survey Design Weights
  id = SEQN,
  weight_mec_2yr = WTMEC2YR,
  survey_psu = SDMVPSU,

```

```

survey_strata      = SDMVSTRA,    # retaining in data incase I want to explore survey weights in an

# Demographics
age_years          = RIDAGEYR,
sex,
race_ethnicity,
marital_status,

# Survival Outcomes
mortality_status   = mortstat,      # 0 = Alive, 1 = Deceased
followup_months    = permth_int,
cause_of_death,

# Primary Exposures (Depression Metrics)
phq9_score         = phq9_sum,
probable_depression, # Binary flag
depression_5bin,    # 5 Bins of the PHQ scale
dsm5_depression    = dsm5_mdd, # Scale 0/1 but based on DSM Scale

# Biometrics & Vitals
bmi                = BMXBMI,
height_cm          = BMXHT,
weight_kg          = BMXWT,
systolic_bp        = BPXSY1,
diastolic_bp       = BPXDI1,
cholesterol_total  = LBXTC,
cholesterol_hdl    = LBDHDD,

# Sleep Habits
sleep_hours        = SLD012,
sleep_category     = sleep_category,
sleep_time_weekday = SLQ300,
wake_time_weekday  = SLQ310,

# Substance Use and Health Behaviours
smoking_status,
age_started_smoking = SMD030,
alcohol_drinks_day  = ALQ130,
used_marijuana,
used_hard_drugs
)

```

Missing data assessment

```

# Inspecting Data for missing-ness
missing_audit <- analysis_clean_df |>

# Calculate missing counts across every single column
summarise(across(everything(), ~ sum(is.na(.x)))) |>

# Pivot to a vertical format for easy viewing
pivot_longer(

```

```

cols = everything(),
names_to = "Variable_Name",
values_to = "Missing_Count"
) |>

# Add percentage metrics relative to the entire eligible cohort size
mutate(
  Total_Cohort_N = nrow(analysis_df),
  Percent_Missing = round((Missing_Count / Total_Cohort_N) * 100, 2)
) |>
# Sort from the most missing to the cleanest structural variables
arrange(desc(Missing_Count))

missing_audit

```

```

## # A tibble: 31 x 4
##   Variable_Name      Missing_Count Total_Cohort_N Percent_Missing
##   <chr>              <int>          <int>          <dbl>
## 1 age_started_smoking      3558            5974            59.6
## 2 alcohol_drinks_day       2604            5974            43.6
## 3 used_marijuana           2559            5974            42.8
## 4 used_hard_drugs          1735            5974            29.0
## 5 phq9_score                850            5974            14.2
## 6 probable_depression       850            5974            14.2
## 7 depression_5bin          850            5974            14.2
## 8 dsm5_depression           850            5974            14.2
## 9 cholesterol_total        592            5974             9.91
## 10 cholesterol_hdl         592            5974             9.91
## # i 21 more rows

```

Baseline Characteristics Table

```

# Prepare and subset data for Table 1
table1_data <- analysis_clean_df |>
  select(
    `Age (Years)` = age_years,
    `Sex` = sex,
    `Race/Ethnicity` = race_ethnicity,
    `Marital Status` = marital_status,
    `Mortality Status` = mortality_status,
    `Follow-up (Months)` = followup_months,
    probable_depression,
    dsm5_depression
  ) |>
  pivot_longer(
    cols = c(probable_depression, dsm5_depression),
    names_to = "depression_metric",
    values_to = "status"
  ) |>
  mutate(
    # Convert to factor and explicitly define level order for tbl_strata

```

```

depression_metric = factor(
  case_when(
    depression_metric == "probable_depression" ~ "Probable Depression (PHQ-9 10+)",
    depression_metric == "dsm5_depression"      ~ "DSM-5 Algorithmic Depression"
  ),
  levels = c(
    "Probable Depression (PHQ-9  $\geq 10$ )",
    "DSM-5 Algorithmic Depression"
  )
),
status = case_when(
  status == 1 ~ "Yes",
  status == 0 ~ "No",
  TRUE       ~ NA_character_
)
) |>
filter(!is.na(status))

# Build Table 1 structured by both metrics side-by-side
table1_demographics <- table1_data |>
tbl_strata(
  strata = depression_metric,
  ~ .x |>
  tbl_summary(
    by = status,
    missing = "ifany",
    missing_text = "Missing",
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_continuous() ~ 1
  ) |>
  add_p(
    test = list(
      all_continuous() ~ "t.test",
      all_categorical() ~ "chisq.test"
    ),
    test.args = list(all_categorical() ~ list(simulate.p.value = TRUE))
  )
) |>
bold_labels() |>
modify_caption("Socio-Demographic Characteristics and Survival Outcomes Stratified by Two Depression Metrics")

table1_demographics

```

Possible Confounders Table

```

# Table 2 for possible confounders
table2_data <- analysis_clean_df |>
select(

```

Table 5: Socio-Demographic Characteristics and Survival Outcomes Stratified by Two Depression Classifications

Characteristic	DSM-5 Algorithmic Depression			NA	
	No N = 4,931 ¹	Yes N = 193 ¹	p-value ²	No N = 4,709 ¹	Yes N =
Age in years at screening	48.2 (18.5)	49.8 (17.7)	0.2	48.2 (18.5)	49.1 (18.5)
Sex			0.011		
Male	2,426 (49%)	77 (40%)		2,339 (50%)	164 (40%)
Female	2,505 (51%)	116 (60%)		2,370 (50%)	251 (60%)
Race/Ethnicity			0.010		
MexicanAmerican	895 (18%)	35 (18%)		861 (18%)	69 (17%)
OtherHispanic	619 (13%)	37 (19%)		586 (12%)	70 (17%)
Non-HispWhite	1,641 (33%)	68 (35%)		1,557 (33%)	152 (37%)
Non-HispBlack	1,055 (21%)	39 (20%)		1,010 (21%)	84 (20%)
Non-HispAsian	532 (11%)	7 (3.6%)		521 (11%)	18 (4.3%)
Other/Multi-Racial	189 (3.8%)	7 (3.6%)		174 (3.7%)	22 (5.3%)
Marital Status			<0.001		
Divorced	498 (11%)	31 (17%)		465 (10%)	64 (16%)
LivingWithPartner	451 (9.6%)	23 (12%)		435 (9.7%)	39 (9.8%)
Married	2,398 (51%)	55 (30%)		2,320 (52%)	133 (33%)
NeverMarried	848 (18%)	48 (26%)		794 (18%)	102 (26%)
Separated	164 (3.5%)	9 (4.9%)		145 (3.2%)	28 (7.0%)
Widowed	333 (7.1%)	19 (10%)		320 (7.1%)	32 (8.0%)
Missing	239	8		230	17
Mortality Status	198 (4.0%)	12 (6.2%)	0.15	183 (3.9%)	27 (6.5%)
Follow-up (Months)	47.5 (8.4)	46.6 (8.8)	0.2	47.6 (8.3)	46.6 (9.1)

¹Mean (SD); n (%)

²Welch Two Sample t-test; Pearson's Chi-squared test with simulated p-value (based on 2000 replicates)

```

`Body Mass Index (BMI)`           = bmi,
`Systolic Blood Pressure (mmHg)`   = systolic_bp,
`Total Cholesterol (mg/dL)`       = cholesterol_total,
`Sleep Duration (Hours)`          = sleep_hours,
`Sleep Category`                  = sleep_category,
`Age started smoking`             = age_started_smoking,
`Smoking Status`                  = smoking_status,
`Alcohol Drinks/Day`              = alcohol_drinks_day,
`History of Marijuana Use`        = used_marijuana,
`History of Hard Drug Use`        = used_hard_drugs,
probable_depression,
dsm5_depression
) |>
pivot_longer(
  cols = c(probable_depression, dsm5_depression),
  names_to = "depression_metric",

```

```

    values_to = "status"
  ) |>
  mutate(
    # Convert to factor and explicitly define level order for tbl_strata
    depression_metric = factor(
      case_when(
        depression_metric == "probable_depression" ~ "Probable Depression (PHQ-9 10+)",
        depression_metric == "dsm5_depression"      ~ "DSM-5 Algorithmic Depression"
      ),
      levels = c(
        "Probable Depression (PHQ-9  $\geq$  10)",
        "DSM-5 Algorithmic Depression"
      )
    ),
    status = case_when(
      status == 1 ~ "Yes",
      status == 0 ~ "No",
      TRUE       ~ NA_character_
    )
  ) |>
  filter(!is.na(status))

# Build Table 2 structured by both metrics side-by-side
table2_confounders <- table2_data |>
  tbl_strata(
    strata = depression_metric,
    ~ .x |>
    tbl_summary(
      by = status,
      missing = "no", # Hides missing counts
      statistic = list(
        all_continuous() ~ "{mean} ({sd})",
        all_categorical() ~ "{n} ({p}%)"
      ),
      type = list(
        `History of Marijuana Use` ~ "categorical",
        `History of Hard Drug Use` ~ "categorical"
      ),
      digits = all_continuous() ~ 1
    ) |>
    add_p(
      test = list(
        all_continuous() ~ "t.test",
        all_categorical() ~ "chisq.test"
      ),
      test.args = list(all_categorical() ~ list(simulate.p.value = TRUE))
    )
  ) |>
  bold_labels() |>
  modify_caption("Clinical Baseline Confounders and Substance Use Behaviours Stratified by Two Depressi")

table2_confounders

```

Table 6: Clinical Baseline Confounders and Substance Use Behaviours Stratified by Two Depression Classifications

Characteristic	DSM-5 Algorithmic Depression			No N = 4
	No N = 4,931 ¹	Yes N = 193 ¹	p-value ²	
Body Mass Index (kg/m**2)	29.4 (7.1)	31.3 (8.0)	0.001	29.4 (7.1)
Systolic: Blood pres (1st rdg) mm Hg	125.1 (18.4)	126.9 (18.9)	0.2	125.1 (18.4)
Total Cholesterol (mg/dL)	189.0 (41.6)	190.6 (42.6)	0.6	188.9 (41.6)
Sleep hours	7.7 (1.5)	8.1 (2.2)	0.045	7.7 (1.5)
Sleep Category			<0.001	
Short (<6 hrs)	393 (8.0%)	25 (13%)		359 (7.7%)
Normal (6-8 hrs)	2,942 (60%)	83 (44%)		2,833 (60%)
Long (>8 hrs)	1,576 (32%)	81 (43%)		1,497 (32%)
Age started smoking cigarettes regularly	22.6 (68.1)	17.4 (6.9)	0.002	22.4 (66.1)
Smoking Status			<0.001	
NeverSmoked	2,923 (59%)	67 (35%)		2,822 (59%)
FormerSmoker	1,124 (23%)	43 (22%)		1,079 (23%)
CurrentSmoker	878 (18%)	83 (43%)		802 (17%)
Alcohol Drinks/Day	2.7 (2.3)	3.8 (3.5)	<0.001	2.7 (2.3)
History of Marijuana Use			<0.001	
No	1,657 (51%)	33 (26%)		1,600 (51%)
Yes	1,619 (49%)	93 (74%)		1,527 (49%)
History of Hard Drug Use			0.003	
No	3,431 (85%)	122 (75%)		3,287 (85%)
Yes	627 (15%)	41 (25%)		586 (15%)

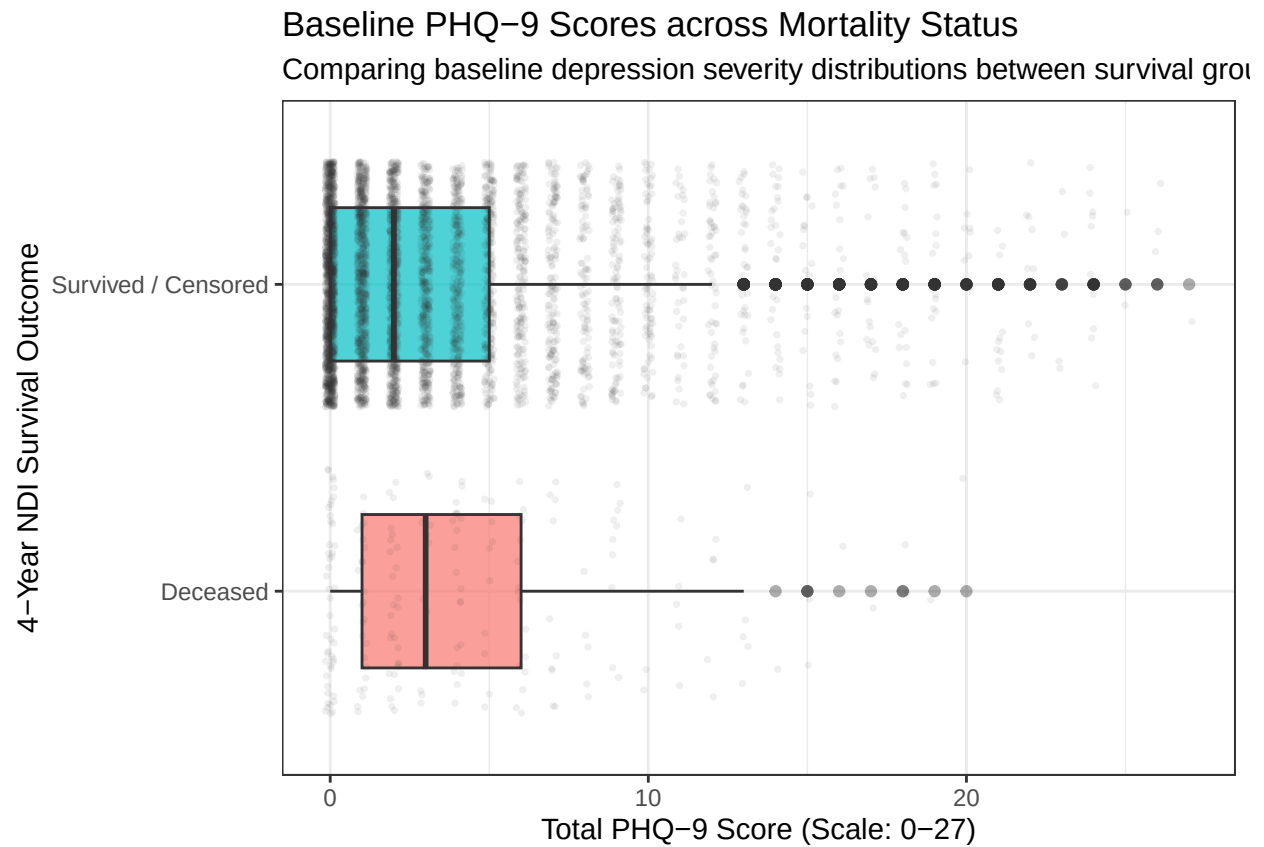
¹Mean (SD); n (%)

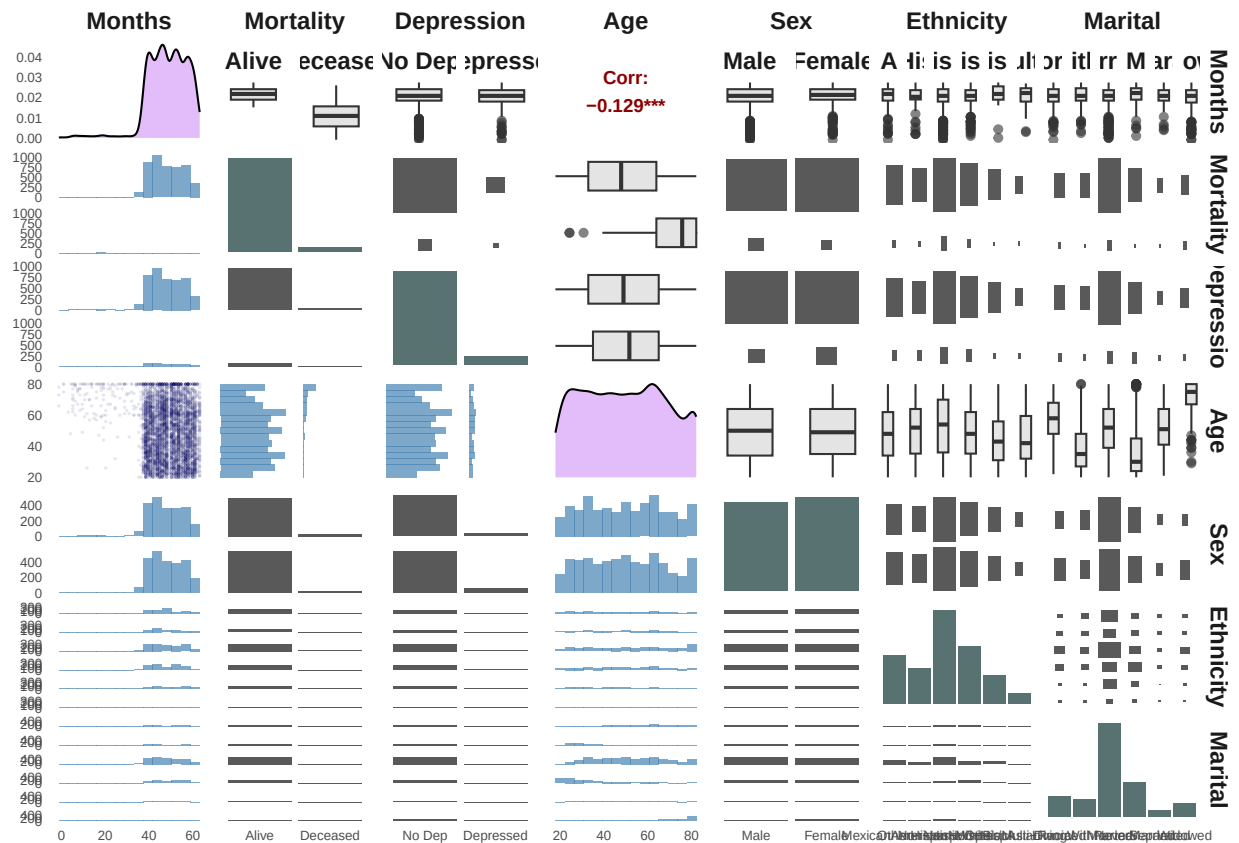
²Welch Two Sample t-test; Pearson's Chi-squared test with simulated p-value (based on 2000 replicates)

Exploratory Analysis

```
#Exploratory Plot
exploratory_plot <- analysis_clean_df |>
  filter(!is.na(phq9_score), !is.na(mortality_status)) |>
  # Convert the 0/1 indicator to clear character labels
  mutate(Survival = ifelse(mortality_status == 1, "Deceased", "Survived / Censored")) |>
  ggplot(aes(y = Survival, x = phq9_score, fill = Survival)) +
  geom_boxplot(alpha = 0.7, outlier.alpha = 0.4, width = 0.5) +
  # Adds a faint point jitter to see where the bulk of the population sits
  geom_jitter(alpha = 0.075, width = 0.15, size = 0.6, color = "grey20") +
  theme_bw() +
  labs(
    title = "Baseline PHQ-9 Scores across Mortality Status",
    subtitle = "Comparing baseline depression severity distributions between survival groups",
    y = "4-Year NDI Survival Outcome",
    x = "Total PHQ-9 Score (Scale: 0-27)"
  )
```

```
) +  
theme(legend.position = "none")  
  
exploratory_plot
```





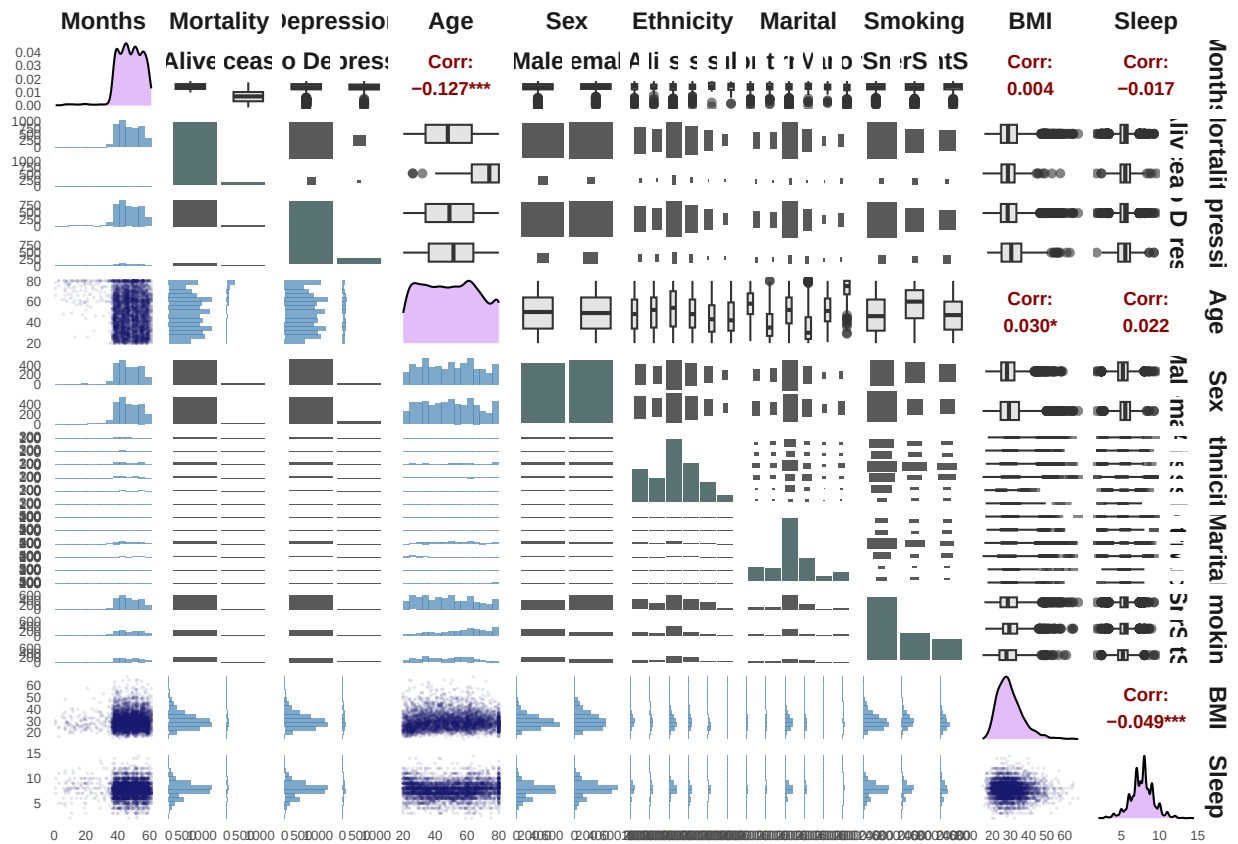
```
pairs_data <- analysis_clean_df |>
  select(
    `Months`      = followup_months,
    `Mortality`   = mortality_status,
    `Depression`  = probable_depression,
    `Age`         = age_years,
    `Sex`         = sex,
    `Ethnicity`   = race_ethnicity,
    `Marital`     = marital_status,
    `Smoking`     = smoking_status,
    `BMI`         = bmi,
    `Sleep`       = sleep_hours
  ) |>
  # Convert binary numeric codes to explicit factors so ggpairs flags them correctly
  mutate(
    Mortality = factor(Mortality, labels = c("Alive", "Deceased")),
    Depression = factor(Depression, labels = c("No Dep", "Depressed"))
  ) |>
  # Drop missing rows
  drop_na()

# Generalized pairs plot with customized sizing to prevent truncation
ggpairs(
  pairs_data,
  lower = list(
    continuous = wrap("points", alpha = 0.1, size = 0.1, color = "midnightblue"),
```

```

    combo = wrap("facethist", bins = 15, fill = "steelblue", alpha = 0.7)
  ),
  upper = list(
    continuous = wrap("cor", size = 2.5, fontface = "bold", color = "darkred"),
    combo = wrap("box", fill = "lightgrey", alpha = 0.6)
  ),
  diag = list(
    continuous = wrap("densityDiag", fill = "purple", alpha = 0.3),
    discrete = wrap("barDiag", fill = "darkslategrey", alpha = 0.8)
  )
) +
theme_minimal(base_size = 8) + # Reduces baseline text size globally to clear space
theme(
  axis.text = element_text(size = 5),                # Shrinks axis number scales
  strip.text = element_text(face = "bold", size = 9), # Makes variable labels clear
  panel.grid.major = element_blank(),                # Strips clutter lines
  panel.grid.minor = element_blank()
)

```



Casual Diagram

```
# Casual Diagram
```

```

depression_dag <- dagitty('dag {
  "Demographics" -> "Depression Exposure"
  "Demographics" -> "All-Cause Mortality"
  "Demographics" -> "Lifestyle Pathways"

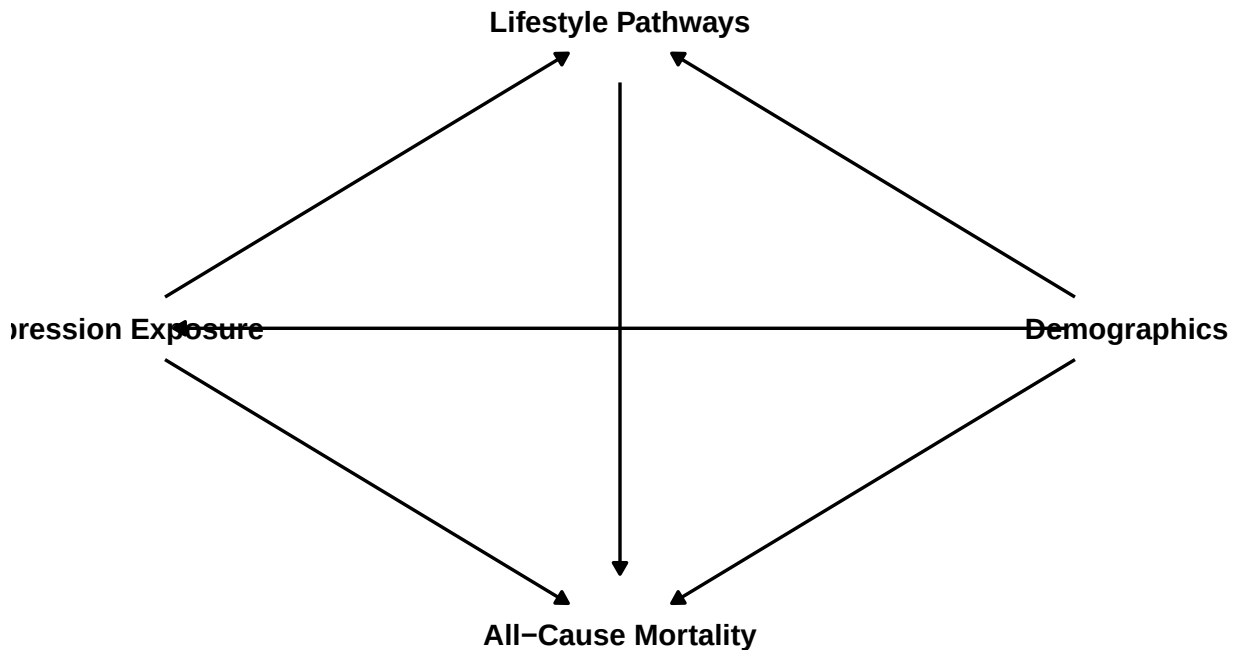
  "Depression Exposure" -> "All-Cause Mortality"
  "Depression Exposure" -> "Lifestyle Pathways"

  "Lifestyle Pathways" -> "All-Cause Mortality"
}')

# Render Doc
ggdag(depression_dag, text_col = "black", node = FALSE) +
  theme_dag() +
  labs(title = "Causal Graph (DAG) for Depression and Long-Term Mortality Profile",
        subtitle = "Mapping Confounders vs. Downstream Behavioural Pathways")

```

Causal Graph (DAG) for Depression and Long-Term Mortality Profile Mapping Confounders vs. Downstream Behavioural Pathways



```

full_mortality_dag <- dagitty('dag {
  "Depression" [exposure]
  "All-Cause Mortality" [outcome]

  "Age" [confounder]
  "Sex" [confounder]
  "Ethnicity" [confounder]

```

```

"Marital Status" [confounder]

"Sleep Duration" [mediator]
"Smoking Status" [mediator]
"Alcohol" [mediator]
"Marijuana Use" [mediator]
"Hard Drug Use" [mediator]
"BMI" [mediator]


"Age" -> "Depression"
"Age" -> "All-Cause Mortality"
"Age" -> "Sleep Duration"
"Age" -> "BMI"


"Sex" -> "Depression"
"Sex" -> "All-Cause Mortality"
"Sex" -> "Smoking Status"
"Sex" -> "Alcohol"
"Sex" -> "Marijuana Use"
"Sex" -> "Hard Drug Use"


"Ethnicity" -> "Depression"
"Ethnicity" -> "All-Cause Mortality"


"Marital Status" -> "Depression"
"Marital Status" -> "All-Cause Mortality"


"Depression" -> "All-Cause Mortality"
"Depression" -> "Sleep Duration"
"Depression" -> "Smoking Status"
"Depression" -> "Alcohol"
"Depression" -> "Marijuana Use"
"Depression" -> "Hard Drug Use"
"Depression" -> "BMI"

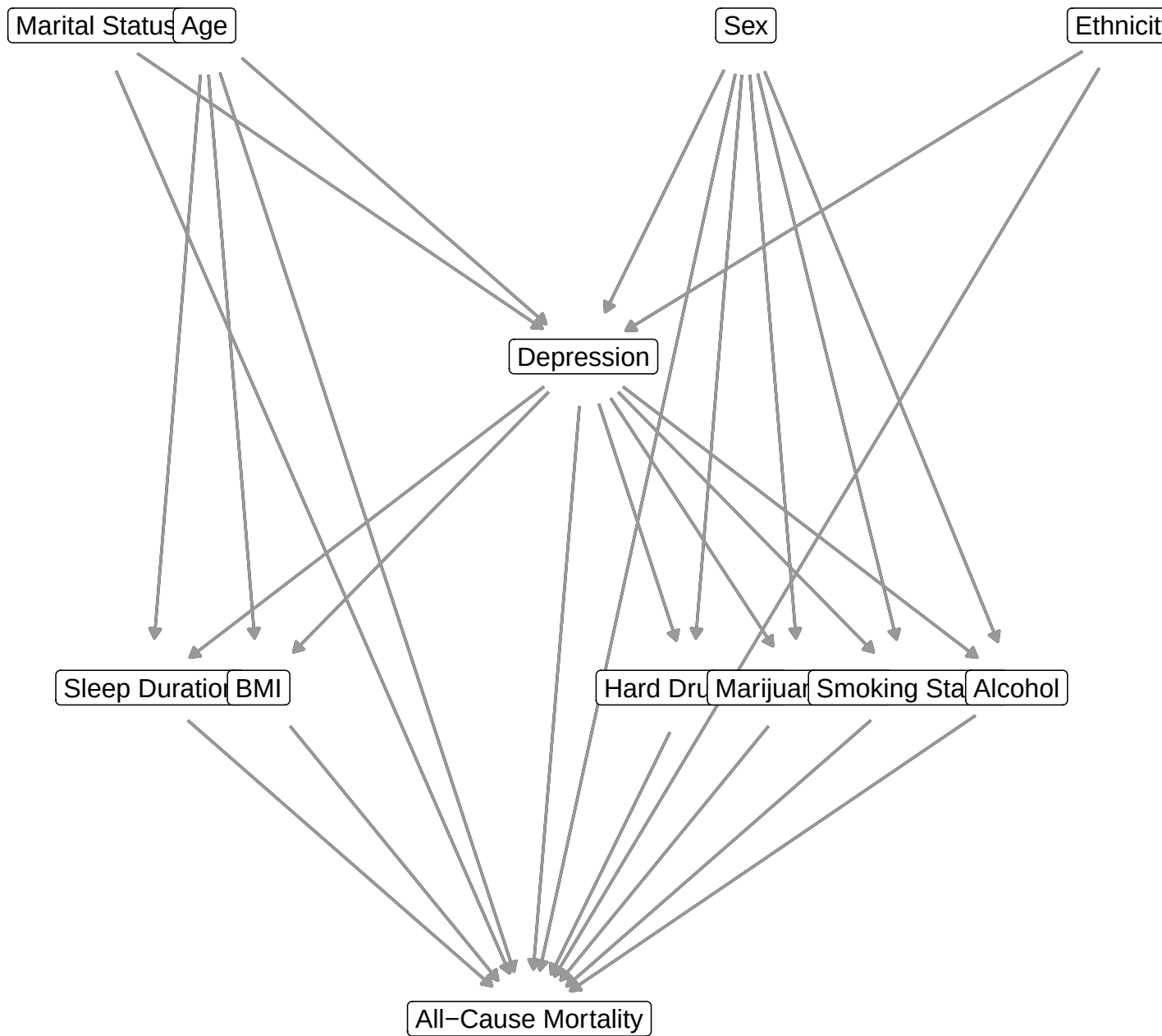

"Sleep Duration" -> "All-Cause Mortality"
"Smoking Status" -> "All-Cause Mortality"
"Alcohol" -> "All-Cause Mortality"
"Marijuana Use" -> "All-Cause Mortality"
"Hard Drug Use" -> "All-Cause Mortality"
"BMI" -> "All-Cause Mortality"
}')

ggdag(full_mortality_dag,
  text = FALSE,          # Turn off default text
  node = FALSE,          # Turn off default labels
  layout = "sugiyama") +
# Draw the arrows first
geom_dag_edges(edge_color = "gray60") +
theme_dag() +
geom_dag_label(
  aes(label = name),

```

```
    color = "black",
    fill = "white",
    label.padding = unit(0.25, "lines")
) +
labs(title = "Directed Acyclic Graph (DAG) of Variable Pathways") +
theme(
  plot.title = element_text(face = "bold", size = 14)
)
```

Directed Acyclic Graph (DAG) of Variable Pathways



Total and Direct Effects

```
adjustmentSets(  
  full_mortality_dag,  
  exposure = "Depression",  
  outcome = "All-Cause Mortality",  
  effect = "total"  
)
```

```
## { Age, Ethnicity, Marital Status, Sex }
```

```
adjustmentSets(  
  full_mortality_dag,  
  exposure = "Depression",  
  outcome = "All-Cause Mortality",  
  effect = "direct"  
)
```

```
## { Age, Alcohol, BMI, Ethnicity, Hard Drug Use, Marijuana Use, Marital  
##   Status, Sex, Sleep Duration, Smoking Status }
```

Condensed Diagram

```
# Condensed Causal Diagram  
condensed_mortality_dag <- dagitty('dag {  
  "Depression" [exposure]  
  "All-Cause Mortality" [outcome]  
  
  "Age" [confounder]  
  "Sex" [confounder]  
  "Ethnicity" [confounder]  
  "Marital Status" [confounder]  
  
  "Sleep Duration" [mediator]  
  "Smoking Status" [mediator]  
  "BMI" [mediator]  
  
  "Age" -> "Depression"  
  "Age" -> "All-Cause Mortality"  
  "Age" -> "Sleep Duration"  
  "Age" -> "BMI"  
  
  "Sex" -> "Depression"  
  "Sex" -> "All-Cause Mortality"  
  "Sex" -> "Smoking Status"  
  
  "Ethnicity" -> "Depression"  
  "Ethnicity" -> "All-Cause Mortality"
```

```

"Marital Status" -> "Depression"
"Marital Status" -> "All-Cause Mortality"

"Depression" -> "All-Cause Mortality"
"Depression" -> "Sleep Duration"
"Depression" -> "Smoking Status"
"Depression" -> "BMI"

"Sleep Duration" -> "All-Cause Mortality"
"Smoking Status" -> "All-Cause Mortality"
"BMI" -> "All-Cause Mortality"
}')

causal_diag_condensed <- ggdag(condensed_mortality_dag,
  text = FALSE,          # Turn off default text
  node = FALSE,         # Turn off default labels
  layout = "sugiyama") +
  # Draw the arrows first
  geom_dag_edges(edge_color = "gray60") +
  theme_dag() +
  geom_dag_label(
    aes(label = name),
    color = "black",
    fill = "white",
    label.size = 0.3,
    label.padding = unit(0.25, "lines")
  ) +
  labs(title = "Directed Acyclic Graph (DAG) of Variable Pathways") +
  theme(
    plot.title = element_text(face = "bold", size = 14)
  )

```

```

## Warning in geom_dag_label(aes(label = name), color = "black", fill = "white", :
## Ignoring unknown parameters: 'label.size'

```

```
causal_diag_condensed
```

Directed Acyclic Graph (DAG) of Variable Pathways



Updated Adjustment Sets

```
adjustmentSets(  
  condensed_mortality_dag,  
  exposure = "Depression",  
  outcome = "All-Cause Mortality",  
  effect = "total"  
)
```

```
## { Age, Ethnicity, Marital Status, Sex }
```

```
adjustmentSets(  
  condensed_mortality_dag,  
  exposure = "Depression",  
  outcome = "All-Cause Mortality",  
  effect = "direct"  
)
```

```
## { Age, BMI, Ethnicity, Marital Status, Sex, Sleep Duration, Smoking  
##   Status }
```

PHQ-9 Depression Models

```
# Raw  
model_raw_effect <- coxph(  
  Surv(followup_months, mortality_status) ~ probable_depression,  
  data = analysis_clean_df  
)  
summary(model_raw_effect)
```

```
## Call:  
## coxph(formula = Surv(followup_months, mortality_status) ~ probable_depression,  
##       data = analysis_clean_df)  
##  
##      n= 5124, number of events= 210  
##      (850 observations deleted due to missingness)  
##  
##              coef exp(coef) se(coef)      z Pr(>|z|)  
## probable_depression 0.5394    1.7150   0.2062 2.616  0.00889 **  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
##              exp(coef) exp(-coef) lower .95 upper .95  
## probable_depression    1.715    0.5831    1.145    2.569  
##  
## Concordance= 0.525  (se = 0.012 )  
## Likelihood ratio test= 5.99  on 1 df,   p=0.01  
## Wald test              = 6.85  on 1 df,   p=0.009  
## Score (logrank) test = 7.01  on 1 df,   p=0.008
```

```

# Total Effect
model_total_effect <- coxph(
  Surv(followup_months, mortality_status) ~ probable_depression +
    age_years + sex + race_ethnicity + marital_status,
  data = analysis_clean_df
)
summary(model_total_effect)

## Call:
## coxph(formula = Surv(followup_months, mortality_status) ~ probable_depression +
##       age_years + sex + race_ethnicity + marital_status, data = analysis_clean_df)
##
##      n= 4877, number of events= 209
##      (1097 observations deleted due to missingness)
##
##              coef exp(coef)  se(coef)      z Pr(>|z|)
## probable_depression      0.592953  1.809323  0.209156  2.835  0.00458
## age_years                0.086860  1.090744  0.006957 12.486 < 2e-16
## sexFemale                -0.876030  0.416433  0.153625 -5.702 1.18e-08
## race_ethnicityOtherHispanic -0.265493  0.766828  0.306740 -0.866  0.38675
## race_ethnicityNon-HispWhite  0.235836  1.265966  0.213866  1.103  0.27015
## race_ethnicityNon-HispBlack  0.033039  1.033590  0.250988  0.132  0.89527
## race_ethnicityNon-HispAsian -0.821343  0.439841  0.487027 -1.686  0.09171
## race_ethnicityOther/Multi-Racial 0.533667  1.705174  0.368979  1.446  0.14808
## marital_statusLivingWithPartner 0.362807  1.437358  0.345862  1.049  0.29418
## marital_statusMarried        -0.380443  0.683559  0.207995 -1.829  0.06739
## marital_statusNeverMarried    -0.266781  0.765841  0.341659 -0.781  0.43490
## marital_statusSeparated       0.231664  1.260697  0.379042  0.611  0.54108
## marital_statusWidowed        0.192306  1.212042  0.230398  0.835  0.40390
##
## probable_depression      **
## age_years                ***
## sexFemale                ***
## race_ethnicityOtherHispanic
## race_ethnicityNon-HispWhite
## race_ethnicityNon-HispBlack
## race_ethnicityNon-HispAsian .
## race_ethnicityOther/Multi-Racial
## marital_statusLivingWithPartner
## marital_statusMarried    .
## marital_statusNeverMarried
## marital_statusSeparated
## marital_statusWidowed
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##              exp(coef) exp(-coef) lower .95 upper .95
## probable_depression      1.8093      0.5527      1.2008      2.7262
## age_years                1.0907      0.9168      1.0760      1.1057
## sexFemale                0.4164      2.4013      0.3082      0.5627
## race_ethnicityOtherHispanic 0.7668      1.3041      0.4203      1.3989
## race_ethnicityNon-HispWhite 1.2660      0.7899      0.8325      1.9252
## race_ethnicityNon-HispBlack 1.0336      0.9675      0.6320      1.6904

```

```
## race_ethnicityNon-HispAsian      0.4398      2.2736      0.1693      1.1425
## race_ethnicityOther/Multi-Racial  1.7052      0.5865      0.8274      3.5143
## marital_statusLivingWithPartner  1.4374      0.6957      0.7297      2.8312
## marital_statusMarried             0.6836      1.4629      0.4547      1.0276
## marital_statusNeverMarried        0.7658      1.3058      0.3920      1.4961
## marital_statusSeparated           1.2607      0.7932      0.5997      2.6500
## marital_statusWidowed             1.2120      0.8251      0.7716      1.9038
##
## Concordance= 0.856 (se = 0.011 )
## Likelihood ratio test= 406 on 13 df, p=<2e-16
## Wald test = 274.8 on 13 df, p=<2e-16
## Score (logrank) test = 401.2 on 13 df, p=<2e-16
```

```
tidy(model_total_effect, exponentiate = TRUE, conf.int = TRUE)
```

```
## # A tibble: 13 x 7
##   term                estimate std.error statistic p.value conf.low conf.high
##   <chr>                <dbl>     <dbl>     <dbl>   <dbl>   <dbl>   <dbl>
## 1 probable_depression  1.81      0.209      2.83 4.58e- 3    1.20    2.73
## 2 age_years           1.09      0.00696    12.5 8.96e-36    1.08    1.11
## 3 sexFemale           0.416      0.154     -5.70 1.18e- 8    0.308    0.563
## 4 race_ethnicityOther~ 0.767      0.307     -0.866 3.87e- 1    0.420    1.40
## 5 race_ethnicityNon-H~ 1.27      0.214      1.10 2.70e- 1    0.832    1.93
## 6 race_ethnicityNon-H~ 1.03      0.251      0.132 8.95e- 1    0.632    1.69
## 7 race_ethnicityNon-H~ 0.440      0.487     -1.69 9.17e- 2    0.169    1.14
## 8 race_ethnicityOther~ 1.71      0.369      1.45 1.48e- 1    0.827    3.51
## 9 marital_statusLivin~ 1.44      0.346      1.05 2.94e- 1    0.730    2.83
## 10 marital_statusMarri~ 0.684      0.208     -1.83 6.74e- 2    0.455    1.03
## 11 marital_statusNever~ 0.766      0.342     -0.781 4.35e- 1    0.392    1.50
## 12 marital_statusSepar~ 1.26      0.379      0.611 5.41e- 1    0.600    2.65
## 13 marital_statusWidow~ 1.21      0.230      0.835 4.04e- 1    0.772    1.90
```

```
# Direct Effect
model_direct_effect <- coxph(
  Surv(followup_months, mortality_status) ~ probable_depression +
    age_years + sex + race_ethnicity + marital_status +
    smoking_status + alcohol_drinks_day + used_marijuana + used_hard_drugs + bmi + sleep_hours,
  data = analysis_clean_df
)
```

```
## Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
## Loglik converged before variable 7,12 ; coefficient may be infinite.
```

```
summary(model_direct_effect)
```

```
## Call:
## coxph(formula = Surv(followup_months, mortality_status) ~ probable_depression +
##   age_years + sex + race_ethnicity + marital_status + smoking_status +
##   alcohol_drinks_day + used_marijuana + used_hard_drugs + bmi +
##   sleep_hours, data = analysis_clean_df)
##
## n= 2291, number of events= 17
```

```

##      (3683 observations deleted due to missingness)
##
##
##              coef  exp(coef)  se(coef)      z
## probable_depression -7.594e-02  9.269e-01  7.826e-01 -0.097
## age_years           9.449e-02  1.099e+00  3.003e-02  3.147
## sexFemale          -8.912e-01  4.102e-01  5.768e-01 -1.545
## race_ethnicityOtherHispanic  6.359e-01  1.889e+00  1.241e+00  0.512
## race_ethnicityNon-HispWhite  3.873e-01  1.473e+00  1.103e+00  0.351
## race_ethnicityNon-HispBlack  8.504e-01  2.341e+00  1.111e+00  0.765
## race_ethnicityNon-HispAsian -1.713e+01  3.645e-08  5.716e+03 -0.003
## race_ethnicityOther/Multi-Racial  5.058e-01  1.658e+00  1.454e+00  0.348
## marital_statusLivingWithPartner -7.295e-01  4.822e-01  7.155e-01 -1.020
## marital_statusMarried -2.195e+00  1.114e-01  7.033e-01 -3.121
## marital_statusNeverMarried -1.037e+00  3.544e-01  7.236e-01 -1.434
## marital_statusSeparated -2.386e+01  4.340e-11  9.527e+04  0.000
## marital_statusWidowed      NA      NA  0.000e+00      NA
## smoking_statusFormerSmoker  3.672e-01  1.444e+00  8.130e-01  0.452
## smoking_statusCurrentSmoker  1.258e+00  3.520e+00  6.755e-01  1.863
## alcohol_drinks_day -5.915e-02  9.426e-01  1.176e-01 -0.503
## used_marijuanaYes  1.911e-02  1.019e+00  6.115e-01  0.031
## used_hard_drugsYes -4.729e-01  6.232e-01  6.208e-01 -0.762
## bmi                6.830e-03  1.007e+00  3.858e-02  0.177
## sleep_hours        7.037e-02  1.073e+00  1.604e-01  0.439
##
##              Pr(>|z|)
## probable_depression  0.92270
## age_years            0.00165 **
## sexFemale            0.12236
## race_ethnicityOtherHispanic  0.60835
## race_ethnicityNon-HispWhite  0.72546
## race_ethnicityNon-HispBlack  0.44401
## race_ethnicityNon-HispAsian  0.99761
## race_ethnicityOther/Multi-Racial  0.72792
## marital_statusLivingWithPartner  0.30794
## marital_statusMarried  0.00180 **
## marital_statusNeverMarried  0.15166
## marital_statusSeparated  0.99980
## marital_statusWidowed      NA
## smoking_statusFormerSmoker  0.65149
## smoking_statusCurrentSmoker  0.06246 .
## alcohol_drinks_day  0.61488
## used_marijuanaYes  0.97507
## used_hard_drugsYes  0.44623
## bmi                0.85949
## sleep_hours        0.66089
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
##              exp(coef) exp(-coef) lower .95 upper .95
## probable_depression  9.269e-01  1.079e+00  0.19992  4.297
## age_years            1.099e+00  9.098e-01  1.03628  1.166
## sexFemale            4.102e-01  2.438e+00  0.13242  1.270
## race_ethnicityOtherHispanic  1.889e+00  5.294e-01  0.16590  21.504
## race_ethnicityNon-HispWhite  1.473e+00  6.789e-01  0.16960  12.793
## race_ethnicityNon-HispBlack  2.341e+00  4.272e-01  0.26524  20.655

```

```
## race_ethnicityNon-HispAsian      3.645e-08  2.744e+07  0.00000      Inf
## race_ethnicityOther/Multi-Racial 1.658e+00  6.030e-01  0.09595     28.662
## marital_statusLivingWithPartner  4.822e-01  2.074e+00  0.11862      1.960
## marital_statusMarried             1.114e-01  8.979e+00  0.02806      0.442
## marital_statusNeverMarried        3.544e-01  2.822e+00  0.08580      1.464
## marital_statusSeparated            4.340e-11  2.304e+10  0.00000      Inf
## marital_statusWidowed             NA         NA         NA         NA
## smoking_statusFormerSmoker        1.444e+00  6.926e-01  0.29338      7.105
## smoking_statusCurrentSmoker       3.520e+00  2.841e-01  0.93662     13.229
## alcohol_drinks_day                9.426e-01  1.061e+00  0.74859      1.187
## used_marijuanaYes                 1.019e+00  9.811e-01  0.30744      3.379
## used_hard_drugsYes                6.232e-01  1.605e+00  0.18458      2.104
## bmi                              1.007e+00  9.932e-01  0.93352      1.086
## sleep_hours                       1.073e+00  9.321e-01  0.78347      1.469
##
## Concordance= 0.899 (se = 0.029 )
## Likelihood ratio test= 47.47 on 19 df,  p=3e-04
## Wald test                = 19.67 on 19 df,  p=0.4
## Score (logrank) test = 59.46 on 19 df,  p=5e-06
```

```
tidy(model_direct_effect, exponentiate = TRUE, conf.int = TRUE)
```

```
## # A tibble: 20 x 7
##   term                estimate std.error statistic p.value conf.low conf.high
##   <chr>                <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 probable_depression  9.27e- 1  7.83e-1 -0.0970   0.923    0.200    4.30
## 2 age_years           1.10e+ 0  3.00e-2  3.15     0.00165  1.04    1.17
## 3 sexFemale           4.10e- 1  5.77e-1 -1.54     0.122    0.132    1.27
## 4 race_ethnicityOthe~ 1.89e+ 0  1.24e+0  0.512    0.608    0.166   21.5
## 5 race_ethnicityNon~~ 1.47e+ 0  1.10e+0  0.351    0.725    0.170   12.8
## 6 race_ethnicityNon~~ 2.34e+ 0  1.11e+0  0.765    0.444    0.265   20.7
## 7 race_ethnicityNon~~ 3.64e- 8  5.72e+3 -0.00300  0.998    0      Inf
## 8 race_ethnicityOthe~ 1.66e+ 0  1.45e+0  0.348    0.728    0.0960  28.7
## 9 marital_statusLivi~ 4.82e- 1  7.15e-1 -1.02     0.308    0.119    1.96
## 10 marital_statusMarr~ 1.11e- 1  7.03e-1 -3.12     0.00180  0.0281   0.442
## 11 marital_statusNeve~ 3.54e- 1  7.24e-1 -1.43     0.152    0.0858   1.46
## 12 marital_statusSepa~ 4.34e-11  9.53e+4 -0.000250 1.000    0      Inf
## 13 marital_statusWido~ NA         0      NA      NA      NA      NA
## 14 smoking_statusForm~ 1.44e+ 0  8.13e-1  0.452    0.651    0.293    7.10
## 15 smoking_statusCurr~ 3.52e+ 0  6.75e-1  1.86     0.0625   0.937   13.2
## 16 alcohol_drinks_day  9.43e- 1  1.18e-1 -0.503    0.615    0.749    1.19
## 17 used_marijuanaYes   1.02e+ 0  6.12e-1  0.0312   0.975    0.307    3.38
## 18 used_hard_drugsYes  6.23e- 1  6.21e-1 -0.762    0.446    0.185    2.10
## 19 bmi                 1.01e+ 0  3.86e-2  0.177    0.859    0.934    1.09
## 20 sleep_hours         1.07e+ 0  1.60e-1  0.439    0.661    0.783    1.47
```

```
#Cleaned Direct Effect
```

```
model_direct_effect_fixed <- coxph(
  Surv(followup_months, mortality_status) ~ probable_depression +
    age_years + sex + race_ethnicity + marital_status +
    smoking_status + bmi + sleep_hours,
  data = analysis_clean_df
)
```

```
summary(model_direct_effect_fixed)
```

```
## Call:
## coxph(formula = Surv(followup_months, mortality_status) ~ probable_depression +
##   age_years + sex + race_ethnicity + marital_status + smoking_status +
##   bmi + sleep_hours, data = analysis_clean_df)
##
##   n= 4805, number of events= 198
##   (1169 observations deleted due to missingness)
##
##               coef exp(coef) se(coef)      z
## probable_depression    0.5127874  1.6699396  0.2157888  2.376
## age_years              0.0866847  1.0905528  0.0075698 11.451
## sexFemale             -0.6580925  0.5178382  0.1627048 -4.045
## race_ethnicityOtherHispanic -0.2595284  0.7714153  0.3162041 -0.821
## race_ethnicityNon-HispWhite  0.2301109  1.2587396  0.2180723  1.055
## race_ethnicityNon-HispBlack -0.0781045  0.9248678  0.2635007 -0.296
## race_ethnicityNon-HispAsian -0.6844701  0.5043574  0.4957718 -1.381
## race_ethnicityOther/Multi-Racial 0.5345279  1.7066424  0.3739627  1.429
## marital_statusLivingWithPartner 0.2953570  1.3436059  0.3474656  0.850
## marital_statusMarried    -0.4014831  0.6693267  0.2137991 -1.878
## marital_statusNeverMarried -0.3375266  0.7135330  0.3552869 -0.950
## marital_statusSeparated    0.1585838  1.1718501  0.3982719  0.398
## marital_statusWidowed      0.1579331  1.1710879  0.2351954  0.671
## smoking_statusFormerSmoker  0.4298926  1.5370924  0.1703188  2.524
## smoking_statusCurrentSmoker 0.7260957  2.0669947  0.2143472  3.387
## bmi                    -0.0009449  0.9990556  0.0123355 -0.077
## sleep_hours              0.0743147  1.0771458  0.0446338  1.665
##
##               Pr(>|z|)
## probable_depression    0.017485 *
## age_years              < 2e-16 ***
## sexFemale              5.24e-05 ***
## race_ethnicityOtherHispanic 0.411782
## race_ethnicityNon-HispWhite 0.291332
## race_ethnicityNon-HispBlack 0.766916
## race_ethnicityNon-HispAsian 0.167397
## race_ethnicityOther/Multi-Racial 0.152900
## marital_statusLivingWithPartner 0.395307
## marital_statusMarried    0.060401 .
## marital_statusNeverMarried 0.342107
## marital_statusSeparated    0.690498
## marital_statusWidowed      0.501904
## smoking_statusFormerSmoker 0.011601 *
## smoking_statusCurrentSmoker 0.000705 ***
## bmi                    0.938943
## sleep_hours              0.095915 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##               exp(coef) exp(-coef) lower .95 upper .95
## probable_depression    1.6699    0.5988    1.0940    2.5491
## age_years              1.0906    0.9170    1.0745    1.1069
## sexFemale              0.5178    1.9311    0.3764    0.7123
```

```
## race_ethnicityOtherHispanic      0.7714      1.2963      0.4151      1.4336
## race_ethnicityNon-HispWhite      1.2587      0.7944      0.8209      1.9300
## race_ethnicityNon-HispBlack      0.9249      1.0812      0.5518      1.5501
## race_ethnicityNon-HispAsian      0.5044      1.9827      0.1909      1.3327
## race_ethnicityOther/Multi-Racial  1.7066      0.5859      0.8200      3.5519
## marital_statusLivingWithPartner  1.3436      0.7443      0.6800      2.6548
## marital_statusMarried            0.6693      1.4940      0.4402      1.0177
## marital_statusNeverMarried       0.7135      1.4015      0.3556      1.4316
## marital_statusSeparated          1.1719      0.8534      0.5369      2.5579
## marital_statusWidowed            1.1711      0.8539      0.7386      1.8569
## smoking_statusFormerSmoker       1.5371      0.6506      1.1008      2.1462
## smoking_statusCurrentSmoker      2.0670      0.4838      1.3580      3.1462
## bmi                              0.9991      1.0009      0.9752      1.0235
## sleep_hours                      1.0771      0.9284      0.9869      1.1756
##
## Concordance= 0.861 (se = 0.011 )
## Likelihood ratio test= 398 on 17 df,  p=<2e-16
## Wald test                = 266.4 on 17 df,  p=<2e-16
## Score (logrank) test = 401.5 on 17 df,  p=<2e-16
```

```
tidy(model_direct_effect_fixed, exponentiate = TRUE, conf.int = TRUE)
```

```
## # A tibble: 17 x 7
##   term                estimate std.error statistic  p.value conf.low conf.high
##   <chr>                <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 probable_depression  1.67      0.216      2.38  1.75e- 2    1.09      2.55
## 2 age_years           1.09      0.00757    11.5  2.31e-30    1.07      1.11
## 3 sexFemale           0.518     0.163     -4.04  5.24e- 5    0.376     0.712
## 4 race_ethnicityOther~ 0.771     0.316     -0.821 4.12e- 1    0.415     1.43
## 5 race_ethnicityNon-H~ 1.26      0.218      1.06  2.91e- 1    0.821     1.93
## 6 race_ethnicityNon-H~ 0.925     0.264     -0.296 7.67e- 1    0.552     1.55
## 7 race_ethnicityNon-H~ 0.504     0.496     -1.38  1.67e- 1    0.191     1.33
## 8 race_ethnicityOther~ 1.71      0.374      1.43  1.53e- 1    0.820     3.55
## 9 marital_statusLivin~ 1.34      0.347      0.850 3.95e- 1    0.680     2.65
## 10 marital_statusMarri~ 0.669     0.214     -1.88  6.04e- 2    0.440     1.02
## 11 marital_statusNever~ 0.714     0.355     -0.950 3.42e- 1    0.356     1.43
## 12 marital_statusSepar~ 1.17      0.398      0.398 6.90e- 1    0.537     2.56
## 13 marital_statusWidow~ 1.17      0.235      0.671 5.02e- 1    0.739     1.86
## 14 smoking_statusForme~ 1.54      0.170      2.52  1.16e- 2    1.10     2.15
## 15 smoking_statusCurre~ 2.07      0.214      3.39  7.05e- 4    1.36     3.15
## 16 bmi                0.999     0.0123    -0.0766 9.39e- 1    0.975     1.02
## 17 sleep_hours        1.08      0.0446      1.66  9.59e- 2    0.987     1.18
```

```
cox.zph(survey_model_raw) # /> plot()
```

```
##               chisq df    p
## probable_depression 0.000385 1 0.98
## GLOBAL              0.000385 1 0.98
```

```
cox.zph(survey_model_total) #/> plot()
```

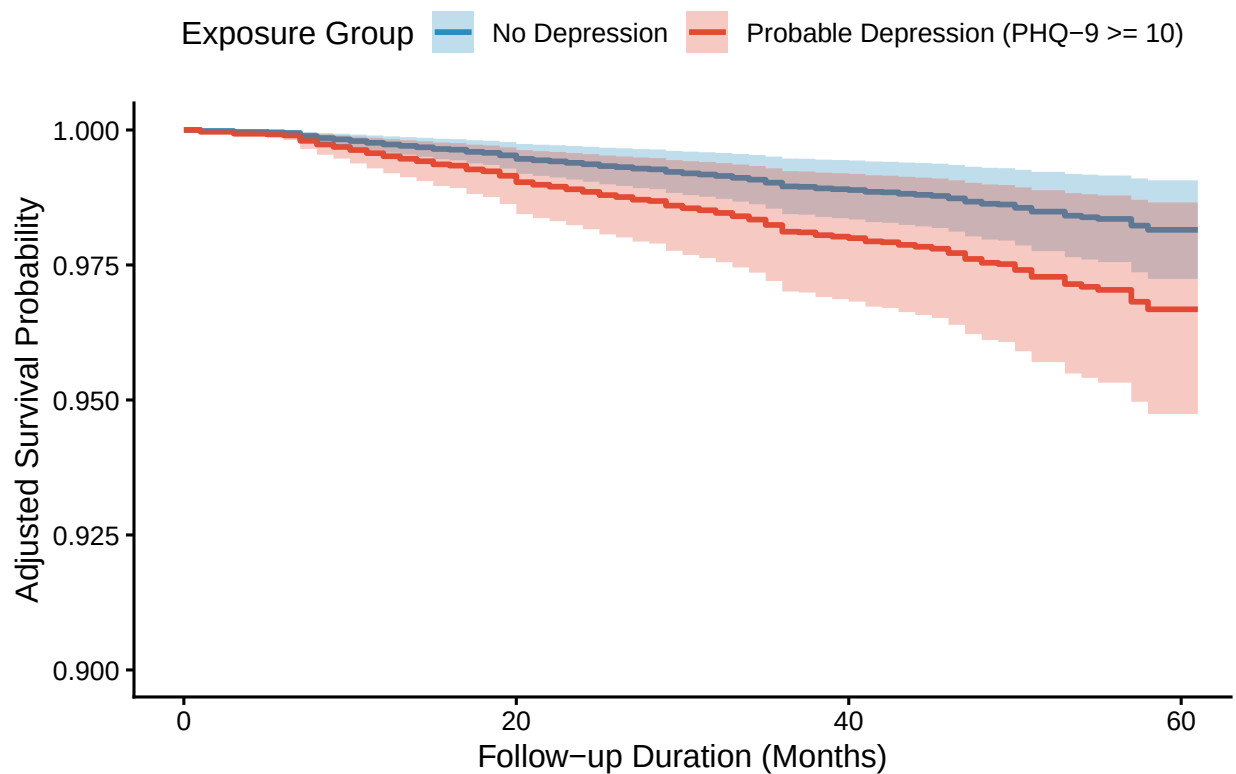
```
##               chisq df    p
```

```
## probable_depression 4.08e-04 1 0.98
## age_years           7.58e-04 1 0.98
## sex                 3.31e-05 1 1.00
## race_ethnicity      6.76e-05 5 1.00
## marital_status      1.39e-04 5 1.00
## GLOBAL              1.41e-03 13 1.00
```

```
cox.zph(survey_model_direct) # /> plot() ---note plots removed from appendix for space saving
```

```
##               chisq df    p
## probable_depression 3.74e-04 1 0.98
## age_years           7.33e-04 1 0.98
## sex                 8.23e-05 1 0.99
## race_ethnicity      8.86e-05 5 1.00
## marital_collapsed   2.38e-06 1 1.00
## smoking_status      2.50e-04 2 1.00
## bmi                 4.49e-04 1 0.98
## sleep_hours         5.21e-05 1 0.99
## GLOBAL              2.84e-03 13 1.00
```

Adjusted Survival Trajectories by Baseline Depression Status



DSM-5 Depression Models

```
# Raw
model_raw_effect_a <- coxph(
  Surv(followup_months, mortality_status) ~ dsm5_depression,
  data = analysis_clean_df
)
summary(model_raw_effect_a)
```

```
## Call:
## coxph(formula = Surv(followup_months, mortality_status) ~ dsm5_depression,
##       data = analysis_clean_df)
##
##      n= 5124, number of events= 210
##      (850 observations deleted due to missingness)
##
##              coef exp(coef) se(coef)      z Pr(>|z|)
## dsm5_depression 0.4593      1.5829   0.2973 1.545   0.122
##
##              exp(coef) exp(-coef) lower .95 upper .95
## dsm5_depression      1.583      0.6317   0.8839      2.835
##
## Concordance= 0.511 (se = 0.008 )
## Likelihood ratio test= 2.09 on 1 df,  p=0.1
## Wald test               = 2.39 on 1 df,  p=0.1
## Score (logrank) test = 2.43 on 1 df,  p=0.1
```

```
# Total
model_total_effect_a <- coxph(
  Surv(followup_months, mortality_status) ~ dsm5_depression +
    age_years + sex + race_ethnicity + marital_status,
  data = analysis_clean_df
)
summary(model_total_effect_a)
```

```
## Call:
## coxph(formula = Surv(followup_months, mortality_status) ~ dsm5_depression +
##       age_years + sex + race_ethnicity + marital_status, data = analysis_clean_df)
##
##      n= 4877, number of events= 209
##      (1097 observations deleted due to missingness)
##
##              coef exp(coef) se(coef)      z Pr(>|z|)
## dsm5_depression 0.430102  1.537414  0.301318  1.427   0.1535
## age_years       0.086803  1.090682  0.006963 12.466 < 2e-16
## sexFemale      -0.861941  0.422341  0.153421 -5.618 1.93e-08
## race_ethnicityOtherHispanic -0.280800  0.755179  0.306867 -0.915  0.3602
## race_ethnicityNon-HispWhite  0.209072  1.232534  0.213496  0.979  0.3274
## race_ethnicityNon-HispBlack  0.002877  1.002882  0.250378  0.011  0.9908
## race_ethnicityNon-HispAsian -0.868748  0.419477  0.486199 -1.787  0.0740
## race_ethnicityOther/Multi-Racial 0.531366  1.701254  0.369354  1.439  0.1503
## marital_statusLivingWithPartner 0.349361  1.418161  0.345829  1.010  0.3124
## marital_statusMarried      -0.407977  0.664994  0.207457 -1.967  0.0492
## marital_statusNeverMarried -0.273318  0.760851  0.341855 -0.800  0.4240
```

```
## marital_statusSeparated      0.253440  1.288449  0.379042  0.669  0.5037
## marital_statusWidowed        0.161897  1.175739  0.230187  0.703  0.4819
##
## dsm5_depression
## age_years                    ***
## sexFemale                    ***
## race_ethnicityOtherHispanic
## race_ethnicityNon-HispWhite
## race_ethnicityNon-HispBlack
## race_ethnicityNon-HispAsian .
## race_ethnicityOther/Multi-Racial
## marital_statusLivingWithPartner
## marital_statusMarried        *
## marital_statusNeverMarried
## marital_statusSeparated
## marital_statusWidowed
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                                exp(coef) exp(-coef) lower .95 upper .95
## dsm5_depression                1.5374      0.6504      0.8517      2.7751
## age_years                      1.0907      0.9169      1.0759      1.1057
## sexFemale                      0.4223      2.3678      0.3127      0.5705
## race_ethnicityOtherHispanic    0.7552      1.3242      0.4139      1.3780
## race_ethnicityNon-HispWhite    1.2325      0.8113      0.8111      1.8730
## race_ethnicityNon-HispBlack    1.0029      0.9971      0.6139      1.6382
## race_ethnicityNon-HispAsian    0.4195      2.3839      0.1618      1.0878
## race_ethnicityOther/Multi-Racial 1.7013      0.5878      0.8248      3.5088
## marital_statusLivingWithPartner 1.4182      0.7051      0.7200      2.7932
## marital_statusMarried          0.6650      1.5038      0.4428      0.9986
## marital_statusNeverMarried     0.7609      1.3143      0.3893      1.4869
## marital_statusSeparated        1.2884      0.7761      0.6130      2.7084
## marital_statusWidowed          1.1757      0.8505      0.7488      1.8461
##
## Concordance= 0.855 (se = 0.011 )
## Likelihood ratio test= 400.9 on 13 df,  p=<2e-16
## Wald test              = 269.6 on 13 df,  p=<2e-16
## Score (logrank) test = 396.6 on 13 df,  p=<2e-16
```

```
tidy(model_total_effect_a, exponentiate = TRUE, conf.int = TRUE)
```

```
## # A tibble: 13 x 7
##   term                estimate std.error statistic  p.value conf.low conf.high
##   <chr>                <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 dsm5_depression      1.54      0.301      1.43  1.53e- 1    0.852    2.78
## 2 age_years            1.09      0.00696    12.5  1.15e-35    1.08     1.11
## 3 sexFemale            0.422     0.153     -5.62  1.93e- 8    0.313    0.571
## 4 race_ethnicityOther~ 0.755     0.307     -0.915  3.60e- 1    0.414    1.38
## 5 race_ethnicityNon-H~ 1.23      0.213      0.979  3.27e- 1    0.811    1.87
## 6 race_ethnicityNon-H~ 1.00      0.250      0.0115 9.91e- 1    0.614    1.64
## 7 race_ethnicityNon-H~ 0.419     0.486     -1.79  7.40e- 2    0.162    1.09
## 8 race_ethnicityOther~ 1.70      0.369      1.44  1.50e- 1    0.825    3.51
## 9 marital_statusLivin~ 1.42      0.346      1.01  3.12e- 1    0.720    2.79
## 10 marital_statusMarri~ 0.665     0.207     -1.97  4.92e- 2    0.443    0.999
```

```
## 11 marital_statusNever~    0.761    0.342    -0.800  4.24e- 1    0.389    1.49
## 12 marital_statusSepar~    1.29    0.379     0.669  5.04e- 1    0.613    2.71
## 13 marital_statusWidow~    1.18    0.230     0.703  4.82e- 1    0.749    1.85
```

```
# Direct
model_direct_effect_fixed_a <- coxph(
  Surv(followup_months, mortality_status) ~ dsm5_depression +
    age_years + sex + race_ethnicity + marital_status +
    smoking_status + bmi + sleep_hours,
  data = analysis_clean_df
)

summary(model_direct_effect_fixed_a)
```

```
## Call:
## coxph(formula = Surv(followup_months, mortality_status) ~ dsm5_depression +
##     age_years + sex + race_ethnicity + marital_status + smoking_status +
##     bmi + sleep_hours, data = analysis_clean_df)
##
##     n= 4805, number of events= 198
##     (1169 observations deleted due to missingness)
##
##               coef exp(coef) se(coef)      z
## dsm5_depression  3.733e-01  1.453e+00  3.048e-01  1.225
## age_years        8.683e-02  1.091e+00  7.575e-03 11.463
## sexFemale       -6.431e-01  5.257e-01  1.624e-01 -3.960
## race_ethnicityOtherHispanic -2.671e-01  7.656e-01  3.160e-01 -0.845
## race_ethnicityNon-HispWhite  2.132e-01  1.238e+00  2.177e-01  0.980
## race_ethnicityNon-HispBlack -9.897e-02  9.058e-01  2.627e-01 -0.377
## race_ethnicityNon-HispAsian -7.159e-01  4.888e-01  4.950e-01 -1.446
## race_ethnicityOther/Multi-Racial 5.365e-01  1.710e+00  3.741e-01  1.434
## marital_statusLivingWithPartner 2.823e-01  1.326e+00  3.473e-01  0.813
## marital_statusMarried -4.199e-01  6.571e-01  2.134e-01 -1.968
## marital_statusNeverMarried -3.358e-01  7.148e-01  3.554e-01 -0.945
## marital_statusSeparated  1.812e-01  1.199e+00  3.980e-01  0.455
## marital_statusWidowed  1.350e-01  1.144e+00  2.356e-01  0.573
## smoking_statusFormerSmoker  4.334e-01  1.543e+00  1.704e-01  2.544
## smoking_statusCurrentSmoker  7.573e-01  2.133e+00  2.134e-01  3.549
## bmi             -3.423e-05  1.000e+00  1.239e-02 -0.003
## sleep_hours      7.611e-02  1.079e+00  4.495e-02  1.693
##
## Pr(>|z|)
## dsm5_depression  0.220690
## age_years        < 2e-16 ***
## sexFemale        7.48e-05 ***
## race_ethnicityOtherHispanic  0.398040
## race_ethnicityNon-HispWhite  0.327273
## race_ethnicityNon-HispBlack  0.706417
## race_ethnicityNon-HispAsian  0.148121
## race_ethnicityOther/Multi-Racial 0.151592
## marital_statusLivingWithPartner 0.416402
## marital_statusMarried  0.049061 *
## marital_statusNeverMarried 0.344780
## marital_statusSeparated 0.648979
## marital_statusWidowed  0.566707
```

```
## smoking_statusFormerSmoker      0.010956 *
## smoking_statusCurrentSmoker      0.000386 ***
## bmi                              0.997796
## sleep_hours                       0.090431 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                                exp(coef) exp(-coef) lower .95 upper .95
## dsm5_depression                1.4525      0.6884      0.7992      2.6400
## age_years                      1.0907      0.9168      1.0746      1.1070
## sexFemale                      0.5257      1.9024      0.3824      0.7226
## race_ethnicityOtherHispanic    0.7656      1.3062      0.4121      1.4224
## race_ethnicityNon-HispWhite    1.2377      0.8080      0.8078      1.8962
## race_ethnicityNon-HispBlack    0.9058      1.1040      0.5412      1.5159
## race_ethnicityNon-HispAsian    0.4888      2.0460      0.1852      1.2896
## race_ethnicityOther/Multi-Racial 1.7100      0.5848      0.8213      3.5601
## marital_statusLivingWithPartner 1.3261      0.7541      0.6713      2.6197
## marital_statusMarried          0.6571      1.5218      0.4325      0.9983
## marital_statusNeverMarried     0.7148      1.3990      0.3562      1.4345
## marital_statusSeparated        1.1986      0.8343      0.5494      2.6151
## marital_statusWidowed          1.1445      0.8738      0.7213      1.8160
## smoking_statusFormerSmoker     1.5425      0.6483      1.1046      2.1540
## smoking_statusCurrentSmoker    2.1326      0.4689      1.4037      3.2399
## bmi                            1.0000      1.0000      0.9760      1.0245
## sleep_hours                    1.0791      0.9267      0.9881      1.1785
##
## Concordance= 0.86 (se = 0.011 )
## Likelihood ratio test= 394.3 on 17 df,  p=<2e-16
## Wald test              = 262.2 on 17 df,  p=<2e-16
## Score (logrank) test = 397.5 on 17 df,  p=<2e-16
```

```
tidy(model_direct_effect_fixed_a, exponentiate = TRUE, conf.int = TRUE)
```

```
## # A tibble: 17 x 7
##   term                estimate std.error statistic  p.value conf.low conf.high
##   <chr>              <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 dsm5_depression      1.45     0.305     1.22    2.21e- 1     0.799     2.64
## 2 age_years            1.09     0.00757   11.5    2.02e-30     1.07     1.11
## 3 sexFemale            0.526    0.162    -3.96    7.48e- 5     0.382     0.723
## 4 race_ethnicityOther~ 0.766    0.316    -0.845   3.98e- 1     0.412     1.42
## 5 race_ethnicityNon-H~ 1.24     0.218     0.980   3.27e- 1     0.808     1.90
## 6 race_ethnicityNon-H~ 0.906    0.263    -0.377   7.06e- 1     0.541     1.52
## 7 race_ethnicityNon-H~ 0.489    0.495    -1.45    1.48e- 1     0.185     1.29
## 8 race_ethnicityOther~ 1.71     0.374     1.43    1.52e- 1     0.821     3.56
## 9 marital_statusLivin~ 1.33     0.347     0.813   4.16e- 1     0.671     2.62
## 10 marital_statusMarri~ 0.657    0.213    -1.97    4.91e- 2     0.433     0.998
## 11 marital_statusNever~ 0.715    0.355    -0.945   3.45e- 1     0.356     1.43
## 12 marital_statusSepar~ 1.20     0.398     0.455   6.49e- 1     0.549     2.62
## 13 marital_statusWidow~ 1.14     0.236     0.573   5.67e- 1     0.721     1.82
## 14 smoking_statusForme~ 1.54     0.170     2.54    1.10e- 2     1.10     2.15
## 15 smoking_statusCurre~ 2.13     0.213     3.55    3.86e- 4     1.40     3.24
## 16 bmi                 1.000    0.0124    -0.00276 9.98e- 1     0.976     1.02
## 17 sleep_hours         1.08     0.0450     1.69    9.04e- 2     0.988     1.18
```

```

new_data <- data.frame(
  dsm5_depression = c(0, 1),
  age_years       = mean(analysis_clean_df$age_years, na.rm = TRUE),
  sex             = "Male", # reference category
  race_ethnicity  = "MexicanAmerican", # reference baseline
  marital_status  = "Married", # reference baseline
  bmi            = mean(analysis_clean_df$bmi, na.rm = TRUE),
  sleep_hours     = mean(analysis_clean_df$sleep_hours, na.rm = TRUE)
)

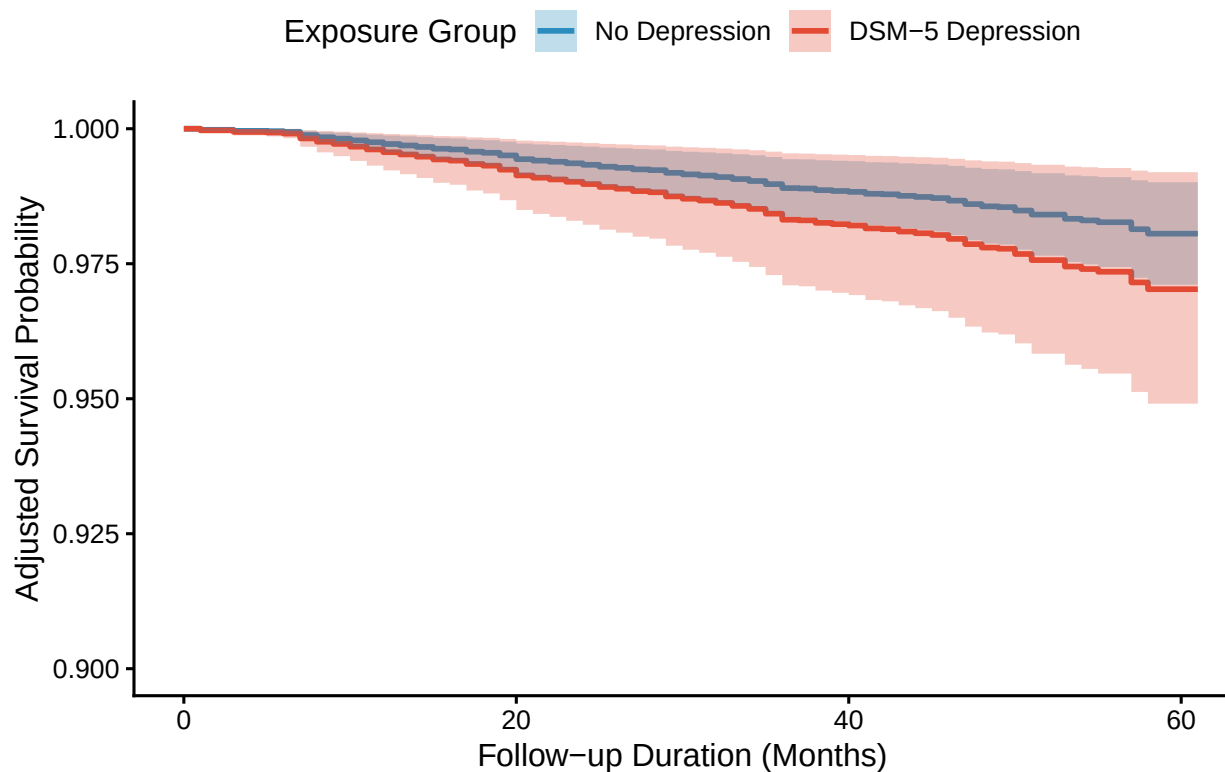
# Comparing survival curves
fit_adjusted <- survfit(model_total_effect_a, newdata = new_data, data = analysis_clean_df)

# Plot
probable_surv_curve_a <- ggsurvplot(
  fit_adjusted,
  data = analysis_clean_df,
  legend.labs = c("No Depression", "DSM-5 Depression"),
  legend.title = "Exposure Group",
  xlab = "Follow-up Duration (Months)",
  ylab = "Adjusted Survival Probability",
  title = "Adjusted Survival Trajectories by Baseline Depression Status",
  ggtheme = theme_classic(base_size = 12),
  conf.int = TRUE,
  censor = FALSE,
  palette = c("#2b8cbe", "#e34a33"),
  ylim = c(0.9, 1)
)

probable_surv_curve_a

```

Adjusted Survival Trajectories by Baseline Depression Status



Survey Weighted Models for PHQ-9 Depression

```
# Survey Weight analysis
analysis_survey_df <- analysis_clean_df |>
  filter(!is.na(weight_mec_2yr) & weight_mec_2yr > 0)

# Allow for adjustment for single PSUs
options(survey.lonely.psu = "adjust")

# Survey Design
nhanes_weighted_design <- svydesign(
  id      = ~survey_psu,      # Primary Sampling Unit variable
  strata  = ~survey_strata,   # Stratum variable
  weights = ~weight_mec_2yr,  # Easeline 2-year examination weights
  nest    = TRUE,            # Nest PSUs inside strata
  data    = analysis_survey_df
)

# Model RAw Effect
survey_model_raw <- svycoxph(
  Surv(followup_months, mortality_status) ~ probable_depression,
  design = nhanes_weighted_design
)
```

```
summary(survey_model_raw)
```

```
## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## svydesign(id = ~survey_psu, strata = ~survey_strata, weights = ~weight_mec_2yr,
##   nest = TRUE, data = analysis_survey_df)

## Call:
## svycoxph(formula = Surv(followup_months, mortality_status) ~
##   probable_depression, design = nhanes_weighted_design)
##
##   n= 5124, number of events= 210
##   (596 observations deleted due to missingness)
##
##               coef exp(coef) se(coef) robust se      z Pr(>|z|)
## probable_depression 0.6004    1.8228   0.2419    0.2677 2.243   0.0249 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##               exp(coef) exp(-coef) lower .95 upper .95
## probable_depression    1.823    0.5486    1.079    3.08
##
## Concordance= 0.529 (se = 0.017 )
## Likelihood ratio test= NA on 1 df,  p=NA
## Wald test              = 5.03 on 1 df,  p=0.02
## Score (logrank) test = NA on 1 df,  p=NA
##
## (Note: the likelihood ratio and score tests assume independence of
##   observations within a cluster, the Wald and robust score tests do not).
```

```
tidy(survey_model_raw, exponentiate = TRUE, conf.int = TRUE)
```

```
## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## svydesign(id = ~survey_psu, strata = ~survey_strata, weights = ~weight_mec_2yr,
##   nest = TRUE, data = analysis_survey_df)
```

```
## # A tibble: 1 x 8
##   term          estimate std.error robust.se statistic p.value conf.low conf.high
##   <chr>          <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 probable_de~    1.82     0.242    0.268     2.24  0.0249     1.08     3.08
```

```
# Model Total Effect
```

```
survey_model_total <- svycoxph(
  Surv(followup_months, mortality_status) ~ probable_depression +
  age_years + sex + race_ethnicity + marital_status,
  design = nhanes_weighted_design
)
```

```
summary(survey_model_total)
```

```

## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## svydesign(id = ~survey_psu, strata = ~survey_strata, weights = ~weight_mec_2yr,
##         nest = TRUE, data = analysis_survey_df)

## Call:
## svycoxph(formula = Surv(followup_months, mortality_status) ~
##         probable_depression + age_years + sex + race_ethnicity +
##         marital_status, design = nhanes_weighted_design)
##
##      n= 4877, number of events= 209
##      (843 observations deleted due to missingness)
##
##               coef exp(coef) se(coef) robust se      z
## probable_depression    0.696027  2.005768  0.246838  0.304821  2.283
## age_years              0.090396  1.094608  0.007642  0.010593  8.533
## sexFemale             -0.700287  0.496443  0.169762  0.145202 -4.823
## race_ethnicityOtherHispanic -0.344144  0.708827  0.592310  0.390926 -0.880
## race_ethnicityNon-HispWhite  0.050462  1.051757  0.361438  0.280387  0.180
## race_ethnicityNon-HispBlack  0.018982  1.019164  0.433567  0.407509  0.047
## race_ethnicityNon-HispAsian -0.686740  0.503214  0.738217  0.338517 -2.029
## race_ethnicityOther/Multi-Racial 0.346718  1.414418  0.508165  0.496343  0.699
## marital_statusLivingWithPartner -0.142102  0.867533  0.395133  0.320316 -0.444
## marital_statusMarried      -0.867886  0.419838  0.222782  0.255101 -3.402
## marital_statusNeverMarried  -0.569093  0.566039  0.393825  0.567684 -1.002
## marital_statusSeparated      0.099356  1.104459  0.435126  0.425705  0.233
## marital_statusWidowed      -0.265908  0.766510  0.257892  0.270239 -0.984
##
##               Pr(>|z|)
## probable_depression    0.022407 *
## age_years              < 2e-16 ***
## sexFemale              1.42e-06 ***
## race_ethnicityOtherHispanic 0.378680
## race_ethnicityNon-HispWhite 0.857173
## race_ethnicityNon-HispBlack 0.962847
## race_ethnicityNon-HispAsian 0.042492 *
## race_ethnicityOther/Multi-Racial 0.484837
## marital_statusLivingWithPartner 0.657310
## marital_statusMarried      0.000669 ***
## marital_statusNeverMarried 0.316111
## marital_statusSeparated      0.815458
## marital_statusWidowed      0.325129
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##               exp(coef) exp(-coef) lower .95 upper .95
## probable_depression    2.0058    0.4986    1.1036    3.6454
## age_years              1.0946    0.9136    1.0721    1.1176
## sexFemale              0.4964    2.0143    0.3735    0.6599
## race_ethnicityOtherHispanic 0.7088    1.4108    0.3294    1.5251
## race_ethnicityNon-HispWhite 1.0518    0.9508    0.6071    1.8221
## race_ethnicityNon-HispBlack 1.0192    0.9812    0.4585    2.2653
## race_ethnicityNon-HispAsian 0.5032    1.9872    0.2592    0.9770
## race_ethnicityOther/Multi-Racial 1.4144    0.7070    0.5347    3.7417
## marital_statusLivingWithPartner 0.8675    1.1527    0.4631    1.6253

```

```
## marital_statusMarried          0.4198      2.3819      0.2546      0.6922
## marital_statusNeverMarried     0.5660      1.7667      0.1861      1.7221
## marital_statusSeparated         1.1045      0.9054      0.4795      2.5440
## marital_statusWidowed          0.7665      1.3046      0.4513      1.3018
##
## Concordance= 0.853 (se = 0.02 )
## Likelihood ratio test= NA on 13 df, p=NA
## Wald test = 624.7 on 13 df, p=<2e-16
## Score (logrank) test = NA on 13 df, p=NA
##
## (Note: the likelihood ratio and score tests assume independence of
## observations within a cluster, the Wald and robust score tests do not).
```

```
tidy(survey_model_total, exponentiate = TRUE, conf.int = TRUE)
```

```
## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## svydesign(id = ~survey_psu, strata = ~survey_strata, weights = ~weight_mec_2yr,
## nest = TRUE, data = analysis_survey_df)
```

```
## # A tibble: 13 x 8
##   term          estimate std.error robust.se statistic  p.value conf.low conf.high
##   <chr>         <dbl>     <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 probable_~    2.01      0.247     0.305     2.28  2.24e- 2    1.10     3.65
## 2 age_years     1.09      0.00764   0.0106     8.53  1.42e-17    1.07     1.12
## 3 sexFemale     0.496     0.170     0.145    -4.82  1.42e- 6    0.373    0.660
## 4 race_ethn~    0.709     0.592     0.391    -0.880 3.79e- 1    0.329    1.53
## 5 race_ethn~    1.05     0.361     0.280     0.180 8.57e- 1    0.607    1.82
## 6 race_ethn~    1.02     0.434     0.408     0.0466 9.63e- 1    0.459    2.27
## 7 race_ethn~    0.503     0.738     0.339    -2.03  4.25e- 2    0.259    0.977
## 8 race_ethn~    1.41     0.508     0.496     0.699 4.85e- 1    0.535    3.74
## 9 marital_s~    0.868     0.395     0.320    -0.444 6.57e- 1    0.463    1.63
## 10 marital_s~   0.420     0.223     0.255    -3.40  6.69e- 4    0.255    0.692
## 11 marital_s~   0.566     0.394     0.568    -1.00  3.16e- 1    0.186    1.72
## 12 marital_s~   1.10     0.435     0.426     0.233 8.15e- 1    0.480    2.54
## 13 marital_s~   0.767     0.258     0.270    -0.984 3.25e- 1    0.451    1.30
```

```
# Model Direct Effect
```

```
# Collapse Marital Status - to reduce dof
```

```
analysis_clean_df <- analysis_clean_df |>
  mutate(
    marital_collapsed = case_when(
      marital_status %in% c("Married", "LivingWithPartner") ~ "Partnered",
      marital_status %in% c("Divorced", "Separated", "Widowed", "NeverMarried") ~ "Unpartnered",
      TRUE ~ NA_character_
    ) |> factor(levels = c("Partnered", "Unpartnered"))
  )
```

```
#Rerun weights and design
```

```
analysis_survey_df <- analysis_clean_df |>
  filter(!is.na(weight_mec_2yr) & weight_mec_2yr > 0)
```

```

nhanes_weighted_design <- svydesign(
  id      = ~survey_psu,      # Primary Sampling Unit variable
  strata  = ~survey_strata,   # Stratum variable
  weights = ~weight_mec_2yr,  # Easeline 2-year examination weights
  nest    = TRUE,             # Nest PSUs inside strata
  data    = analysis_survey_df
)

# No NAs
nhanes_direct_design <- subset(
  nhanes_weighted_design,
  !is.na(followup_months) & !is.na(mortality_status) &
  !is.na(probable_depression) & !is.na(smoking_status) &
  !is.na(bmi) & !is.na(sleep_hours) & !is.na(marital_collapsed)
)

survey_model_direct <- svycoxph(
  Surv(followup_months, mortality_status) ~ probable_depression +
    age_years + sex + race_ethnicity + marital_collapsed +
    smoking_status + bmi + sleep_hours,
  design = nhanes_direct_design
)

summary(survey_model_direct)

## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## subset(nhanes_weighted_design, !is.na(followup_months) & !is.na(mortality_status) &
##      !is.na(probable_depression) & !is.na(smoking_status) & !is.na(bmi) &
##      !is.na(sleep_hours) & !is.na(marital_collapsed))

## Call:
## svycoxph(formula = Surv(followup_months, mortality_status) ~
##      probable_depression + age_years + sex + race_ethnicity +
##      marital_collapsed + smoking_status + bmi + sleep_hours,
##      design = nhanes_direct_design)
##
##      n= 4805, number of events= 198
##
##               coef exp(coef)  se(coef) robust se      z
## probable_depression    0.603106  1.827786  0.259888  0.316992  1.903
## age_years              0.088494  1.092528  0.007644  0.011855  7.465
## sexFemale             -0.556113  0.573434  0.180620  0.152665 -3.643
## race_ethnicityOtherHispanic -0.301917  0.739399  0.618814  0.368447 -0.819
## race_ethnicityNon-HispWhite  0.032111  1.032632  0.374060  0.263089  0.122
## race_ethnicityNon-HispBlack -0.081622  0.921620  0.456939  0.380070 -0.215
## race_ethnicityNon-HispAsian -0.515140  0.597417  0.761947  0.314611 -1.637
## race_ethnicityOther/Multi-Racial 0.328953  1.389513  0.522913  0.463260  0.710
## marital_collapsedUnpartnered  0.636299  1.889475  0.175571  0.195927  3.248
## smoking_statusFormerSmoker   0.453623  1.574004  0.194401  0.185681  2.443
## smoking_statusCurrentSmoker  0.654685  1.924537  0.258003  0.390258  1.678

```

```
## bmi 0.022015 1.022259 0.013023 0.011418 1.928
## sleep_hours 0.051018 1.052342 0.058521 0.073492 0.694
## Pr(>|z|)
## probable_depression 0.05709 .
## age_years 8.33e-14 ***
## sexFemale 0.00027 ***
## race_ethnicityOtherHispanic 0.41254
## race_ethnicityNon-HispWhite 0.90286
## race_ethnicityNon-HispBlack 0.82996
## race_ethnicityNon-HispAsian 0.10155
## race_ethnicityOther/Multi-Racial 0.47765
## marital_collapsedUnpartnered 0.00116 **
## smoking_statusFormerSmoker 0.01456 *
## smoking_statusCurrentSmoker 0.09343 .
## bmi 0.05384 .
## sleep_hours 0.48756
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## exp(coef) exp(-coef) lower .95 upper .95
## probable_depression 1.8278 0.5471 0.9820 3.4021
## age_years 1.0925 0.9153 1.0674 1.1182
## sexFemale 0.5734 1.7439 0.4251 0.7734
## race_ethnicityOtherHispanic 0.7394 1.3524 0.3591 1.5223
## race_ethnicityNon-HispWhite 1.0326 0.9684 0.6166 1.7294
## race_ethnicityNon-HispBlack 0.9216 1.0850 0.4376 1.9412
## race_ethnicityNon-HispAsian 0.5974 1.6739 0.3225 1.1068
## race_ethnicityOther/Multi-Racial 1.3895 0.7197 0.5604 3.4450
## marital_collapsedUnpartnered 1.8895 0.5292 1.2870 2.7741
## smoking_statusFormerSmoker 1.5740 0.6353 1.0938 2.2649
## smoking_statusCurrentSmoker 1.9245 0.5196 0.8956 4.1354
## bmi 1.0223 0.9782 0.9996 1.0454
## sleep_hours 1.0523 0.9503 0.9112 1.2154
##
## Concordance= 0.852 (se = 0.019 )
## Likelihood ratio test= NA on 13 df, p=NA
## Wald test = 1197 on 13 df, p=<2e-16
## Score (logrank) test = NA on 13 df, p=NA
##
## (Note: the likelihood ratio and score tests assume independence of
## observations within a cluster, the Wald and robust score tests do not).
```

```
tidy(survey_model_direct, exponentiate = TRUE, conf.int = TRUE)
```

```
## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## subset(nhanes_weighted_design, !is.na(followup_months) & !is.na(mortality_status) &
## !is.na(probable_depression) & !is.na(smoking_status) & !is.na(bmi) &
## !is.na(sleep_hours) & !is.na(marital_collapsed))

## # A tibble: 13 x 8
## term estimate std.error robust.se statistic p.value conf.low conf.high
## <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
```

```
## 1 probable_~ 1.83 0.260 0.317 1.90 5.71e- 2 0.982 3.40
## 2 age_years 1.09 0.00764 0.0119 7.46 8.33e-14 1.07 1.12
## 3 sexFemale 0.573 0.181 0.153 -3.64 2.70e- 4 0.425 0.773
## 4 race_ethn~ 0.739 0.619 0.368 -0.819 4.13e- 1 0.359 1.52
## 5 race_ethn~ 1.03 0.374 0.263 0.122 9.03e- 1 0.617 1.73
## 6 race_ethn~ 0.922 0.457 0.380 -0.215 8.30e- 1 0.438 1.94
## 7 race_ethn~ 0.597 0.762 0.315 -1.64 1.02e- 1 0.322 1.11
## 8 race_ethn~ 1.39 0.523 0.463 0.710 4.78e- 1 0.560 3.45
## 9 marital_c~ 1.89 0.176 0.196 3.25 1.16e- 3 1.29 2.77
## 10 smoking_s~ 1.57 0.194 0.186 2.44 1.46e- 2 1.09 2.26
## 11 smoking_s~ 1.92 0.258 0.390 1.68 9.34e- 2 0.896 4.14
## 12 bmi 1.02 0.0130 0.0114 1.93 5.38e- 2 1.000 1.05
## 13 sleep_hou~ 1.05 0.0585 0.0735 0.694 4.88e- 1 0.911 1.22
```

```
cox.zph(survey_model_raw) # /> plot()
```

```
##                chisq df    p
## probable_depression 0.000385 1 0.98
## GLOBAL              0.000385 1 0.98
```

```
cox.zph(survey_model_total) # /> plot()
```

```
##                chisq df    p
## probable_depression 4.08e-04 1 0.98
## age_years          7.58e-04 1 0.98
## sex                3.31e-05 1 1.00
## race_ethnicity     6.76e-05 5 1.00
## marital_status     1.39e-04 5 1.00
## GLOBAL             1.41e-03 13 1.00
```

```
cox.zph(survey_model_direct) # /> plot() ---note plots removed from appendix for space saving
```

```
##                chisq df    p
## probable_depression 3.74e-04 1 0.98
## age_years          7.33e-04 1 0.98
## sex                8.23e-05 1 0.99
## race_ethnicity     8.86e-05 5 1.00
## marital_collapsed  2.38e-06 1 1.00
## smoking_status     2.50e-04 2 1.00
## bmi                4.49e-04 1 0.98
## sleep_hours        5.21e-05 1 0.99
## GLOBAL             2.84e-03 13 1.00
```

Survey Weighted Models for DSM-5

```
# Survey Weight analysis - DSM5
# Model Total Effect

# Model Raw Effect
survey_model_raw_a <- svycoxph(
```

```

Surv(followup_months, mortality_status) ~ dsm5_depression,
design = nhanes_weighted_design
)

summary(survey_model_raw_a)

## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## svydesign(id = ~survey_psu, strata = ~survey_strata, weights = ~weight_mec_2yr,
##         nest = TRUE, data = analysis_survey_df)

## Call:
## svycoxph(formula = Surv(followup_months, mortality_status) ~
##          dsm5_depression, design = nhanes_weighted_design)
##
##      n= 5124, number of events= 210
##      (596 observations deleted due to missingness)
##
##              coef exp(coef) se(coef) robust se      z Pr(>|z|)
## dsm5_depression 0.6889      1.9916   0.3280   0.4736 1.455   0.146
##
##              exp(coef) exp(-coef) lower .95 upper .95
## dsm5_depression      1.992      0.5021   0.7872   5.038
##
## Concordance= 0.518 (se = 0.015 )
## Likelihood ratio test= NA on 1 df,  p=NA
## Wald test              = 2.12 on 1 df,  p=0.1
## Score (logrank) test = NA on 1 df,  p=NA
##
## (Note: the likelihood ratio and score tests assume independence of
##       observations within a cluster, the Wald and robust score tests do not).

tidy(survey_model_raw_a, exponentiate = TRUE, conf.int = TRUE)

## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## svydesign(id = ~survey_psu, strata = ~survey_strata, weights = ~weight_mec_2yr,
##         nest = TRUE, data = analysis_survey_df)

## # A tibble: 1 x 8
##   term          estimate std.error robust.se statistic p.value conf.low conf.high
##   <chr>          <dbl>     <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
## 1 dsm5_depres~      1.99      0.328    0.474      1.45   0.146   0.787   5.04

survey_model_total_a <- svycoxph(
  Surv(followup_months, mortality_status) ~ dsm5_depression +
    age_years + sex + race_ethnicity + marital_status,
  design = nhanes_weighted_design
)

summary(survey_model_total_a)

```

```
## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## svydesign(id = ~survey_psu, strata = ~survey_strata, weights = ~weight_mec_2yr,
##         nest = TRUE, data = analysis_survey_df)

## Call:
## svycoxph(formula = Surv(followup_months, mortality_status) ~
##         dsm5_depression + age_years + sex + race_ethnicity + marital_status,
##         design = nhanes_weighted_design)
##
##      n= 4877, number of events= 209
##      (843 observations deleted due to missingness)
##
##              coef exp(coef) se(coef) robust se      z
## dsm5_depression    0.767651  2.154698  0.331821  0.516038  1.488
## age_years          0.090320  1.094525  0.007659  0.010713  8.431
## sexFemale          -0.670789  0.511305  0.168918  0.145943 -4.596
## race_ethnicityOtherHispanic -0.379359  0.684300  0.592450  0.374415 -1.013
## race_ethnicityNon-HispWhite  0.025253  1.025575  0.360805  0.260117  0.097
## race_ethnicityNon-HispBlack -0.014563  0.985542  0.432959  0.389324 -0.037
## race_ethnicityNon-HispAsian -0.727669  0.483034  0.737407  0.332333 -2.190
## race_ethnicityOther/Multi-Racial 0.343140  1.409366  0.508478  0.496167  0.692
## marital_statusLivingWithPartner -0.151377  0.859524  0.395172  0.315581 -0.480
## marital_statusMarried -0.900004  0.406568  0.221743  0.247406 -3.638
## marital_statusNeverMarried -0.594124  0.552046  0.393284  0.561839 -1.057
## marital_statusSeparated  0.102336  1.107755  0.434603  0.412229  0.248
## marital_statusWidowed -0.307290  0.735437  0.257649  0.264166 -1.163
##
## Pr(>|z|)
## dsm5_depression    0.136860
## age_years          < 2e-16 ***
## sexFemale          4.3e-06 ***
## race_ethnicityOtherHispanic 0.310964
## race_ethnicityNon-HispWhite 0.922659
## race_ethnicityNon-HispBlack 0.970161
## race_ethnicityNon-HispAsian 0.028555 *
## race_ethnicityOther/Multi-Racial 0.489200
## marital_statusLivingWithPartner 0.631457
## marital_statusMarried 0.000275 ***
## marital_statusNeverMarried 0.290301
## marital_statusSeparated 0.803941
## marital_statusWidowed 0.244729
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##              exp(coef) exp(-coef) lower .95 upper .95
## dsm5_depression    2.1547    0.4641    0.7837    5.9243
## age_years          1.0945    0.9136    1.0718    1.1177
## sexFemale          0.5113    1.9558    0.3841    0.6806
## race_ethnicityOtherHispanic 0.6843    1.4613    0.3285    1.4254
## race_ethnicityNon-HispWhite 1.0256    0.9751    0.6160    1.7076
## race_ethnicityNon-HispBlack 0.9855    1.0147    0.4595    2.1138
## race_ethnicityNon-HispAsian 0.4830    2.0702    0.2518    0.9265
## race_ethnicityOther/Multi-Racial 1.4094    0.7095    0.5329    3.7270
## marital_statusLivingWithPartner 0.8595    1.1634    0.4631    1.5954
```

```
## marital_statusMarried          0.4066      2.4596      0.2503      0.6603
## marital_statusNeverMarried      0.5520      1.8114      0.1835      1.6604
## marital_statusSeparated          1.1078      0.9027      0.4938      2.4850
## marital_statusWidowed           0.7354      1.3597      0.4382      1.2343
##
## Concordance= 0.852 (se = 0.02 )
## Likelihood ratio test= NA on 13 df,  p=NA
## Wald test                      = 797.2 on 13 df,  p=<2e-16
## Score (logrank) test = NA on 13 df,  p=NA
##
## (Note: the likelihood ratio and score tests assume independence of
## observations within a cluster, the Wald and robust score tests do not).
```

```
tidy(survey_model_total_a, exponentiate = TRUE, conf.int = TRUE)
```

```
## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## svydesign(id = ~survey_psu, strata = ~survey_strata, weights = ~weight_mec_2yr,
## nest = TRUE, data = analysis_survey_df)
```

```
## # A tibble: 13 x 8
##   term          estimate std.error robust.se statistic  p.value conf.low conf.high
##   <chr>         <dbl>     <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 dsm5_depr~    2.15     0.332     0.516     1.49  1.37e- 1    0.784    5.92
## 2 age_years     1.09     0.00766    0.0107     8.43  3.43e-17    1.07    1.12
## 3 sexFemale     0.511    0.169     0.146    -4.60  4.30e- 6    0.384    0.681
## 4 race_ethn~    0.684    0.592     0.374    -1.01  3.11e- 1    0.329    1.43
## 5 race_ethn~    1.03    0.361     0.260     0.0971 9.23e- 1    0.616    1.71
## 6 race_ethn~    0.986    0.433     0.389    -0.0374 9.70e- 1    0.459    2.11
## 7 race_ethn~    0.483    0.737     0.332    -2.19  2.86e- 2    0.252    0.927
## 8 race_ethn~    1.41    0.508     0.496     0.692  4.89e- 1    0.533    3.73
## 9 marital_s~    0.860    0.395     0.316    -0.480  6.31e- 1    0.463    1.60
## 10 marital_s~   0.407    0.222     0.247    -3.64  2.75e- 4    0.250    0.660
## 11 marital_s~   0.552    0.393     0.562    -1.06  2.90e- 1    0.184    1.66
## 12 marital_s~   1.11    0.435     0.412     0.248  8.04e- 1    0.494    2.49
## 13 marital_s~   0.735    0.258     0.264    -1.16  2.45e- 1    0.438    1.23
```

```
# Model Direct Effect

# Collapse Marital Status - to reduce dof
analysis_clean_df <- analysis_clean_df |>
  mutate(
    marital_collapsed = case_when(
      marital_status %in% c("Married", "LivingWithPartner") ~ "Partnered",
      marital_status %in% c("Divorced", "Separated", "Widowed", "NeverMarried") ~ "Unpartnered",
      TRUE ~ NA_character_
    ) |> factor(levels = c("Partnered", "Unpartnered"))
  )

#Rerun weights and design
analysis_survey_df <- analysis_clean_df |>
  filter(!is.na(weight_mec_2yr) & weight_mec_2yr > 0)
```

```

nhanes_weighted_design <- svydesign(
  id      = ~survey_psu,      # Primary Sampling Unit variable
  strata  = ~survey_strata,   # Stratum variable
  weights = ~weight_mec_2yr,  # Easeline 2-year examination weights
  nest    = TRUE,             # Nest PSUs inside strata
  data    = analysis_survey_df
)

# No NAs
nhanes_direct_design <- subset(
  nhanes_weighted_design,
  !is.na(followup_months) & !is.na(mortality_status) &
  !is.na(probable_depression) & !is.na(smoking_status) &
  !is.na(bmi) & !is.na(sleep_hours) & !is.na(marital_collapsed)
)

survey_model_direct_a <- svycoxph(
  Surv(followup_months, mortality_status) ~ dsm5_depression +
    age_years + sex + race_ethnicity + marital_collapsed +
    smoking_status + bmi + sleep_hours,
  design = nhanes_direct_design
)

summary(survey_model_direct_a)

## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## subset(nhanes_weighted_design, !is.na(followup_months) & !is.na(mortality_status) &
##      !is.na(probable_depression) & !is.na(smoking_status) & !is.na(bmi) &
##      !is.na(sleep_hours) & !is.na(marital_collapsed))

## Call:
## svycoxph(formula = Surv(followup_months, mortality_status) ~
##      dsm5_depression + age_years + sex + race_ethnicity + marital_collapsed +
##      smoking_status + bmi + sleep_hours, design = nhanes_direct_design)
##
##      n= 4805, number of events= 198
##
##              coef exp(coef)  se(coef) robust se      z
## dsm5_depression    0.632234  1.881810  0.346811  0.523455  1.208
## age_years          0.088193  1.092199  0.007629  0.011844  7.446
## sexFemale         -0.532787  0.586967  0.179797  0.149342 -3.568
## race_ethnicityOtherHispanic -0.315293  0.729575  0.618604  0.352050 -0.896
## race_ethnicityNon-HispWhite  0.016992  1.017137  0.373833  0.248825  0.068
## race_ethnicityNon-HispBlack -0.104237  0.901012  0.456368  0.369726 -0.282
## race_ethnicityNon-HispAsian -0.544955  0.579868  0.761419  0.315145 -1.729
## race_ethnicityOther/Multi-Racial 0.332753  1.394803  0.523539  0.467383  0.712
## marital_collapsedUnpartnered 0.644664  1.905347  0.175475  0.192366  3.351
## smoking_statusFormerSmoker   0.459009  1.582505  0.194380  0.179592  2.556
## smoking_statusCurrentSmoker  0.678014  1.969962  0.257550  0.401610  1.688
## bmi                    0.022649  1.022907  0.013057  0.012128  1.867

```

```
## sleep_hours          0.051277  1.052614  0.058769  0.075363  0.680
##                      Pr(>|z|)
## dsm5_depression      0.227120
## age_years            9.6e-14 ***
## sexFemale            0.000360 ***
## race_ethnicityOtherHispanic 0.370471
## race_ethnicityNon-HispWhite 0.945557
## race_ethnicityNon-HispBlack 0.777998
## race_ethnicityNon-HispAsian 0.083770 .
## race_ethnicityOther/Multi-Racial 0.476496
## marital_collapsedUnpartnered 0.000804 ***
## smoking_statusFormerSmoker 0.010593 *
## smoking_statusCurrentSmoker 0.091365 .
## bmi                  0.061841 .
## sleep_hours          0.496252
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                      exp(coef) exp(-coef) lower .95 upper .95
## dsm5_depression      1.8818      0.5314      0.6745      5.2498
## age_years            1.0922      0.9156      1.0671      1.1178
## sexFemale            0.5870      1.7037      0.4380      0.7866
## race_ethnicityOtherHispanic 0.7296      1.3707      0.3659      1.4546
## race_ethnicityNon-HispWhite 1.0171      0.9832      0.6246      1.6565
## race_ethnicityNon-HispBlack 0.9010      1.1099      0.4365      1.8597
## race_ethnicityNon-HispAsian 0.5799      1.7245      0.3127      1.0754
## race_ethnicityOther/Multi-Racial 1.3948      0.7169      0.5581      3.4862
## marital_collapsedUnpartnered 1.9053      0.5248      1.3069      2.7779
## smoking_statusFormerSmoker 1.5825      0.6319      1.1130      2.2502
## smoking_statusCurrentSmoker 1.9700      0.5076      0.8966      4.3282
## bmi                  1.0229      0.9776      0.9989      1.0475
## sleep_hours          1.0526      0.9500      0.9081      1.2202
##
## Concordance= 0.852 (se = 0.019 )
## Likelihood ratio test= NA on 13 df,  p=NA
## Wald test              = 2160 on 13 df,  p=<2e-16
## Score (logrank) test = NA on 13 df,  p=NA
##
## (Note: the likelihood ratio and score tests assume independence of
## observations within a cluster, the Wald and robust score tests do not).
```

```
tidy(survey_model_direct_a, exponentiate = TRUE, conf.int = TRUE)
```

```
## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## subset(nhanes_weighted_design, !is.na(followup_months) & !is.na(mortality_status) &
##      !is.na(probable_depression) & !is.na(smoking_status) & !is.na(bmi) &
##      !is.na(sleep_hours) & !is.na(marital_collapsed))

## # A tibble: 13 x 8
##   term          estimate std.error robust.se statistic  p.value conf.low conf.high
##   <chr>         <dbl>     <dbl>     <dbl>     <dbl>    <dbl>   <dbl>   <dbl>
## 1 dsm5_depr~    1.88      0.347     0.523     1.21    2.27e- 1    0.675    5.25
```

```
## 2 age_years      1.09    0.00763    0.0118    7.45    9.60e-14    1.07    1.12
## 3 sexFemale      0.587    0.180    0.149   -3.57    3.60e- 4    0.438    0.787
## 4 race_ethn~     0.730    0.619    0.352   -0.896    3.70e- 1    0.366    1.45
## 5 race_ethn~     1.02    0.374    0.249    0.0683    9.46e- 1    0.625    1.66
## 6 race_ethn~     0.901    0.456    0.370   -0.282    7.78e- 1    0.437    1.86
## 7 race_ethn~     0.580    0.761    0.315   -1.73    8.38e- 2    0.313    1.08
## 8 race_ethn~     1.39    0.524    0.467    0.712    4.76e- 1    0.558    3.49
## 9 marital_c~     1.91    0.175    0.192    3.35    8.04e- 4    1.31    2.78
## 10 smoking_s~    1.58    0.194    0.180    2.56    1.06e- 2    1.11    2.25
## 11 smoking_s~    1.97    0.258    0.402    1.69    9.14e- 2    0.897    4.33
## 12 bmi           1.02    0.0131    0.0121    1.87    6.18e- 2    0.999    1.05
## 13 sleep_hou~    1.05    0.0588    0.0754    0.680    4.96e- 1    0.908    1.22
```

```
cox.zph(survey_model_raw_a) # /> plot()
```

```
##                chisq df    p
## dsm5_depression 3.94e-05  1 0.99
## GLOBAL          3.94e-05  1 0.99
```

```
cox.zph(survey_model_total_a) # /> plot()
```

```
##                chisq df    p
## dsm5_depression 4.30e-05  1 0.99
## age_years       7.44e-04  1 0.98
## sex             3.36e-05  1 1.00
## race_ethnicity  6.65e-05  5 1.00
## marital_status  1.35e-04  5 1.00
## GLOBAL          1.09e-03 13 1.00
```

```
cox.zph(survey_model_direct_a) # /> plot() ---note plots removed from appendix for space saving
```

```
##                chisq df    p
## dsm5_depression 3.65e-05  1 1.00
## age_years       7.22e-04  1 0.98
## sex             8.38e-05  1 0.99
## race_ethnicity  8.71e-05  5 1.00
## marital_collapsed 1.89e-06  1 1.00
## smoking_status  2.44e-04  2 1.00
## bmi             4.42e-04  1 0.98
## sleep_hours     5.30e-05  1 0.99
## GLOBAL          2.36e-03 13 1.00
```

Hazard Ratio Summaries

```
# Extract results from all models
hr_raw    <- tidy(model_raw_effect,    exponentiate = TRUE, conf.int = TRUE) |>
  filter(term == "probable_depression") |>
  mutate(Model = "Unadjusted (Raw)", Type = "Unweighted")

hr_total  <- tidy(model_total_effect,  exponentiate = TRUE, conf.int = TRUE) |>
```

```

  filter(term == "probable_depression") |>
  mutate(Model = "Total Effect",      Type = "Unweighted")

hr_direct <- tidy(model_direct_effect_fixed, exponentiate = TRUE, conf.int = TRUE) |>
  filter(term == "probable_depression") |>
  mutate(Model = "Direct Effect",      Type = "Unweighted")

#####
# Stricter DSM-5 Criteria
hr_raw_a <- tidy(model_raw_effect_a,      exponentiate = TRUE, conf.int = TRUE) |>
  filter(term == "dsm5_depression") |>
  mutate(Model = "Unadjusted (Raw)", Type = "Unweighted")

hr_total_a <- tidy(model_total_effect_a,  exponentiate = TRUE, conf.int = TRUE) |>
  filter(term == "dsm5_depression") |>
  mutate(Model = "Total Effect",      Type = "Unweighted")

hr_direct_a <- tidy(model_direct_effect_fixed_a, exponentiate = TRUE, conf.int = TRUE) |>
  filter(term == "dsm5_depression") |>
  mutate(Model = "Direct Effect",      Type = "Unweighted")

#####
# Survey Results
hr_raw_svy <- tidy(survey_model_raw, exponentiate = TRUE, conf.int = TRUE) |>
  filter(term == "probable_depression") |>
  mutate(Model = "Unadjusted (Raw)", Type = "Survey-Weighted")

## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## svydesign(id = ~survey_psu, strata = ~survey_strata, weights = ~weight_mec_2yr,
##          nest = TRUE, data = analysis_survey_df)

hr_total_svy <- tidy(survey_model_total,  exponentiate = TRUE, conf.int = TRUE) |>
  filter(term == "probable_depression") |>
  mutate(Model = "Total Effect", Type = "Survey-Weighted")

## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## svydesign(id = ~survey_psu, strata = ~survey_strata, weights = ~weight_mec_2yr,
##          nest = TRUE, data = analysis_survey_df)

hr_direct_svy <- tidy(survey_model_direct, exponentiate = TRUE, conf.int = TRUE) |>
  filter(term == "probable_depression") |>
  mutate(Model = "Direct Effect", Type = "Survey-Weighted")

## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## subset(nhanes_weighted_design, !is.na(followup_months) & !is.na(mortality_status) &
##      !is.na(probable_depression) & !is.na(smoking_status) & !is.na(bmi) &
##      !is.na(sleep_hours) & !is.na(marital_collapsed))

```

```
#####
# DSM5
hr_raw_svy_a <- tidy(survey_model_raw_a, exponentiate = TRUE, conf.int = TRUE) |>
  filter(term == "dsm5_depression") |>
  mutate(Model = "Unadjusted (Raw)", Type = "Survey-Weighted")

## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## svydesign(id = ~survey_psu, strata = ~survey_strata, weights = ~weight_mec_2yr,
##          nest = TRUE, data = analysis_survey_df)

hr_total_svy_a <- tidy(survey_model_total_a, exponentiate = TRUE, conf.int = TRUE) |>
  filter(term == "dsm5_depression") |>
  mutate(Model = "Total Effect", Type = "Survey-Weighted")

## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## svydesign(id = ~survey_psu, strata = ~survey_strata, weights = ~weight_mec_2yr,
##          nest = TRUE, data = analysis_survey_df)

hr_direct_svy_a <- tidy(survey_model_direct_a, exponentiate = TRUE, conf.int = TRUE) |>
  filter(term == "dsm5_depression") |>
  mutate(Model = "Direct Effect", Type = "Survey-Weighted")

## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## subset(nhanes_weighted_design, !is.na(followup_months) & !is.na(mortality_status) &
##      !is.na(probable_depression) & !is.na(smoking_status) & !is.na(bmi) &
##      !is.na(sleep_hours) & !is.na(marital_collapsed))

# Combine everything together
compare_df <- bind_rows(hr_raw, hr_total, hr_direct,
                        hr_raw_a, hr_total_a, hr_direct_a,
                        hr_raw_svy, hr_total_svy, hr_direct_svy,
                        hr_raw_svy_a, hr_total_svy_a, hr_direct_svy_a) |>
  mutate(
    Model = factor(Model, levels = c("Direct Effect", "Total Effect", "Unadjusted (Raw)")),
    Type = factor(Type, levels = c("Unweighted", "Survey-Weighted"))
  )
```

Hazard Ratio Visualisations

```
# Visualise
facet_names <- c(
  "probable_depression" = "PHQ-9 Definition (Screening Threshold  $\geq 10$ )",
  "dsm5_depression"      = "DSM-5 Definition (Algorithmic Method)"
)

# Grouped Forest Plot
```

```

results_plot <- compare_df |>
  mutate(
    # Force 'probable_depression' to be first
    term = factor(term, levels = c("probable_depression", "dsm5_depression"))
  ) |>
  ggplot(aes(x = estimate, y = Model, color = Type, shape = Type)) +
  geom_vline(xintercept = 1, linetype = "dashed", color = "darkgrey", linewidth = 0.6) +

  # Using position_dodge to bump the unweighted and weighted intervals away from overlapping
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), linewidth = 0.8, width = 0.2,
    position = position_dodge(width = 0.4)) +
  geom_point(size = 3, position = position_dodge(width = 0.4)) +

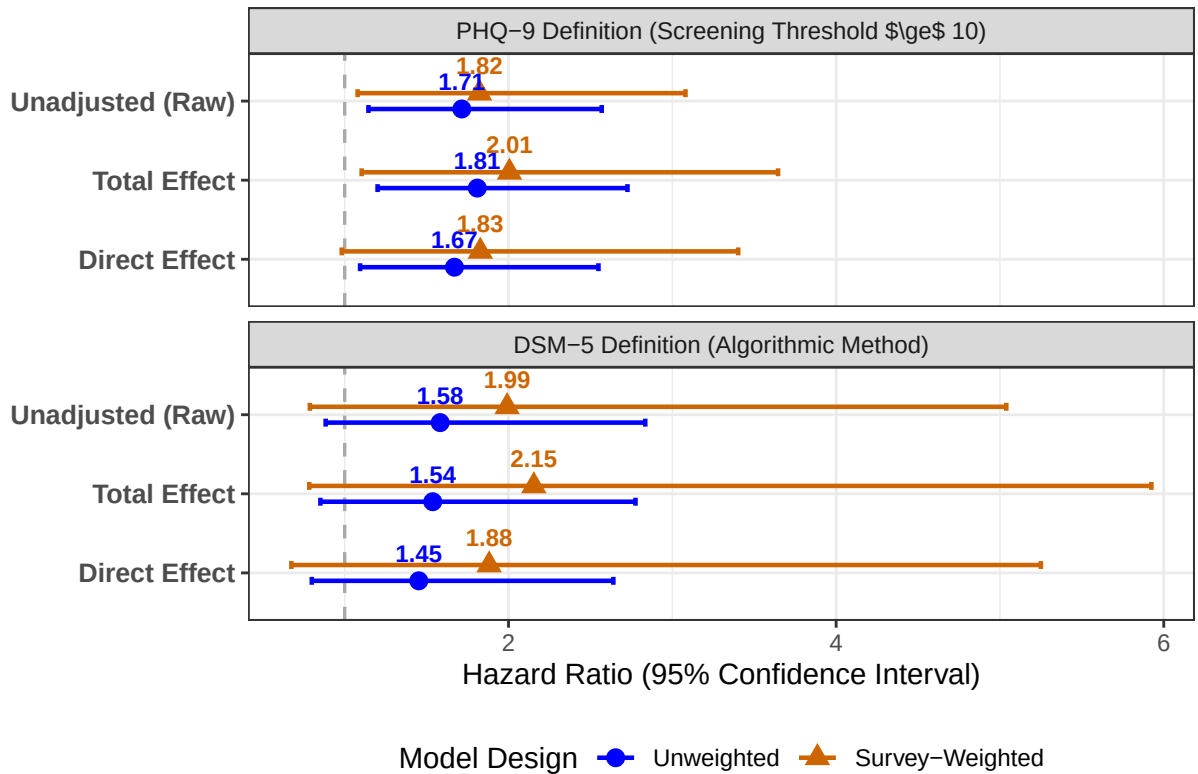
  # Add exact text tags above each estimate line
  geom_text(
    aes(label = sprintf("%.2f", estimate)),
    vjust = -1.1,
    size = 3,
    fontface = "bold",
    position = position_dodge(width = 0.4),
    show.legend = FALSE
  ) +
  scale_color_manual(values = c("blue", "darkorange3")) +
  scale_shape_manual(values = c(16, 17)) + # Circle for unweighted, Triangle for survey

  theme_bw() +
  facet_wrap(~term, ncol=1, labeller = as_labeller(facet_names)) +
  expand_limits(y = c(0.6, 3.5)) +
  labs(
    title = "Sensitivity Analysis: Sampling Weight Impact on Hazards",
    x = "Hazard Ratio (95% Confidence Interval)",
    y = "",
    color = "Model Design",
    shape = "Model Design"
  ) +
  theme(
    plot.title = element_text(face = "bold"),
    axis.text.y = element_text(face = "bold", size = 10),
    legend.position = "bottom"
  )

results_plot

```

Sensitivity Analysis: Sampling Weight Impact on Hazards



Tables for Results

```
# Tables for results
table_data <- compare_df |>
  mutate(
    # Clean up names
    Indicator = case_when(
      term == "probable_depression" ~ "PHQ-9 Depression",
      term == "dsm5_depression" ~ "DSM-5 Depression"
    ),
    # Combine HR and CI into standard format: X.XX (X.XX, X.XX)
    HR_CI = sprintf("%.2f (%.2f, %.2f)", estimate, conf.low, conf.high),
    # Format p-values
    P_Value = ifelse(p.value < 0.001, "<0.001", sprintf("%.3f", p.value))
  ) |>
  select(Type, Indicator, Model, HR_CI, P_Value)

# Split into two separate dataframes
unweighted_df <- table_data |> filter(Type == "Unweighted") |> select(-Type)
weighted_df <- table_data |> filter(Type == "Survey-Weighted") |> select(-Type)
```